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# Exploring parsimonious principles that unify active portfolio selection (II): validation

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This research unifies *active* portfolio selection through (1) *Subjective Allocation Rule* (SAR)-based allocation and (2) *Minimum Tracking Error* (MTE)-guided mimicking within a Bayesian Allocation Framework that integrates qualitative and quantitative beliefs into alpha-generating portfolios. Portfolio selection is modelled as *view-driven, feedback-responsive* rationality that reconciles *normative* and *descriptive* paradigms, with diverse investor behaviours emerging as *local optima* shaped by cognition, skill, and subjectivity. Empirical results show that when *subjectivity* aligns with predictive *skill*, portfolios outperform mean-variance benchmarks. This second paper focuses on the *empirical validation*.

*Keywords*: Biases in portfolio theory validation; Factor mimicking and hedging; Forecasting skill and error; Ambiguity and subjectivity; Monte Carlo; Portfolio performance

*JEL Classifications*: C38, C61, C63, D81, G11, G14, G41

## 1. Introduction

Despite significant advances in portfolio theory, diverse investor portfolio-selection (PS) behaviours – such as *equal weighting*, *factor ranking*, *portfolio algebra*, *volatility-managed portfolios*, *hedging*, and *portable alpha* – remain insufficiently unified by Markowitz (1952) portfolio theory (MPT) or its extensions. Cheung (2024a, 'Paper (I)') demonstrates that two efficiency principles – the *Subjective Allocation Rule* (SAR) (Proposition 4.3, Paper (I)) and *Minimum Tracking Error* (MTE) (Proposition 4.1) – suffice to span a comprehensive, unified active PS framework: Bayesian Allocation Framework. This Paper (II) empirically validates SAR and MTE, demonstrating their efficiency, alpha (outperformance)-generating capacity, and cognitive friendliness through portfolio analytics and backtests, while identifying key performance drivers.

The first principle, *SAR*, allocates based on the view signal-to-noise ratio: higher ratios imply stronger conviction and larger tilts from the benchmark. Paper (I) demonstrates that SAR (1) translates investor beliefs (or predictions) into outperformance proportionally to their forecasting skill‡ (Lemma 4.2), (2) leverages qualitative information (Proposition 4.2), and (3) assists correlation-agnostic investors (Lemma 4.5), thereby (4) effectively utilising investor subjectivity to outperform (Proposition 4.3). This paper develops novel methods, as explained below, to isolate and evaluate these mechanisms, assessing whether SAR efficiently transforms *skill* and *subjectivity* into portfolio gains while simplifying active portfolio selection.

The second principle, *MTE*, optimises mimicking or hedging of target market variables by minimising tracking variances. Paper (I) shows that MTE produces efficient (5) factor mimicking (FM) (Hypothesis 4.1) and (6) security-specific mimicking (SSM) (Hypotheses 4.2) (i.e. portable alpha) portfolios, with the latter hedged against common factor risks. This paper empirically tests these mechanisms to determine whether MTE harnesses broader information sources for effective active portfolio selection.

Lemma 4.2 of Paper (I) establishes that, in modern asset management – where analysts and alpha models produce predictions – an effective PS model should serve as the efficient conduit, converting their forecasting *skill* into portfolio alpha, rather than competing with them to generate alpha independently. Comparing PS models via profitability 'horse races' conflates their skill-to-performance transition efficiency with

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‡ See Subsection 3.2.2 for the technical definition of *skill*.

the quality of input forecasts, resulting in **joint tests** that obscure each model's intrinsic merit. For example, poor forecasts can make a good PS model underperform, while a skilled investor can make an inefficient model appear effective. To isolate transition efficiency, this paper introduces a **skill-controlled view simulation** (SVS) approach to testing how forecasting skill translates into portfolio outperformance (mechanism (1) above).

SAR allows *subjective* adjustments to mean and variance, with different treatments of identical information reflecting real-world portfolio behaviours. This unifying property provides two advantages: (i) an impartial basis for comparing diverse PS models (behaviours), and (ii) a novel **comparative subjectivity testing** (CST) framework for directly assessing how investor subjectivity influences performance (mechanisms (2), (3) and (4) above). Although prior studies have examined subjectivity – in its representation (e.g. Smimou *et al.* 2008), incorporation into optimisation (e.g. Hasuike and Katagiri 2010, Boyle *et al.* 2012), endogenous information choices (e.g. Van Nieuwerburgh and Veldkamp 2010), and investor perceptions (e.g. Calvo-Pardo *et al.* 2022) – its economic value remains unclear. Through CST, this research contributes new empirical evidence by evaluating *subjectivity as a potential performance-enhancing factor*.

Separately, MTE's effectiveness is assessed using both factor and security-specific views. For *factor views*, the relationships between MTE-based factor-mimicking (FM) portfolios and their target factors (mechanism (5)) are examined. While Cheung (2013) and Cheung and Mittal (2009) apply MTE-based factor mimicking to sectoral and simulated factors respectively, this paper offers a more comprehensive analysis encompassing both sectoral and fundamental factors through analytical inspection and out-of-sample (OOS) correlation backtests using a multifactor model fitted to historical data. For *security-specific views*, MTE-based SSM portfolios ('portable alphas') are evaluated against their target securities or baskets ('themes') and the broader market (mechanism (6)). Unlike Cheung (2013), which focuses on single-stock SSM portfolio weights, this paper examines both single-stock and basket-level views through analytical exposures and out-of-sample correlations.

The paper is structured as follows: Section 2 presents the data and the multifactor model for estimating historical covariances and factor time series. Section 3 evaluates SAR across mechanisms (1)-(4), introducing the PS model evaluation methodology, establishing a fair comparison framework, and presenting CST for subjectivity testing and SVS to resolve the joint-test issue. Sections 4 and 5 assess MTE's effectiveness through FM (mechanism (5)) and SSM (mechanism (6)), respectively. Finally, Section 6 discusses how the Bayesian Allocation Framework, spanning SAR and MTE, addresses the seven portfolio optimisation challenges identified by Salo *et al.* (2024).

This paper contributes by: (1) empirically validating SAR and MTE as principles for *active* portfolio selection; (2) designing methods to resolve the joint-test problem in PS model evaluation and quantify the impact of subjectivity on performance; (3) demonstrating optimisation with subjective, non-quantitative information; and (4) identifying the key drivers of active portfolio performance.

## 2. Data and the multifactor model

This paper uses historical variance estimates for portfolio optimisation and model-estimated factor time series† to test broad investor beliefs. For this purpose, a linear, cross-sectional multifactor risk model with $f$ factors is estimated:

$$\widehat{\boldsymbol{\Sigma}} = \mathbf{B}\widehat{\boldsymbol{\Sigma}}_F\mathbf{B}^{\mathrm{T}} + \widehat{\boldsymbol{\Sigma}}_\xi \tag{1}$$

where $\widehat{\boldsymbol{\Sigma}}$ is the $[n \times n]$ covariance matrix of security returns‡; $\mathbf{B}$ is the $[n \times f]$ matrix of factor loadings (exposures); $\widehat{\boldsymbol{\Sigma}}_F$ is the $[f \times f]$ covariance matrix of factor returns; and $\widehat{\boldsymbol{\Sigma}}_\xi$ denotes the $[n \times n]$ covariance matrix of security-specific returns, which are assumed to be independent of each other and of factor returns, following standard factor-modelling conventions.

This model is fitted monthly to the S&P 500 index (SPX) universe from 01 February 2013 to 31 October 2023 (129 months), using S&P Capital IQ daily close prices and quarterly fundamental data. It is technically similar to the Barra family of cross-sectional multifactor risk models but includes 15 sectoral and fundamental factors§ and has a short-term nature.¶ An indicative part of the factor loading matrix $\mathbf{B}$ is shown in Table 1. Sectoral factor loadings (exposures) are binary, with '1' indicating sector membership and '0' otherwise. Fundamental factor loadings are represented by $z$-score normalised stock characteristics.

## 3. Validating SAR: *Skill* and *Subjectivity* impact

This section evaluates how varying mean forecasting skill and subjective treatments of the covariance matrix affect out-of-sample (OOS) performance of view portfolios, measured by Sharpe ratio (SR), Jensen's information ratio (JIR), and other metrics. The results validate Lemmas 4.2 (SAR Alpha Generation with Orthogonal Views) and 4.5 (Orthogonal Representation of Correlated Views), as well as Propositions 4.2 (Feasibility of Pareto Exposure Improvements) and 4.3 (Principle 2: SAR-Based Allocation) from Paper (I).

### 3.1. *Setup*

Assume an initial capital of US$1 million, investable only in the SPX universe of $n = 448$ stocks‖ or a treasury bond

† Factors in this paper refer to those generated by a multifactor model rather than predefined baskets derived from Fama and French (1992, 1993) rankings. To distinguish the two, the latter is referred to as factor 'baskets' or 'themes'.
‡ Unless specified otherwise, returns in this paper refer to excess returns
§ Three criteria guide factor selection: (1) factors must be meaningful for factor-based portfolio analysis, so PCA is avoided; (2) a $k$ ($> 1$)-factor model is generally more accurate than sample-based or single-index approaches to covariance estimation; and (3) too many factors may cause overfitting. On balance, 15 factors are chosen.
¶ Two-month rolling windows of daily data are used to ensure responsiveness to factor movements.
‖ Due to index rebalancing, only 448 of the 500 SPX constituents passed data quality screening.

Table 1. The factor loading matrix.

| | Ind | H Care | IT | Cons D | Fina | Mat | R Est | Util | Comm | Cons S | Energy | BP | DY | Size | Ret_3M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MMM | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.6413 | 2.1671 | − 0.3790 | − 0.6816 |
| AOS | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.5476 | − 0.2591 | 1.1339 | 0.1098 |
| ABT | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.4754 | − 0.1349 | − 1.5324 | 0.3004 |
| ABBV | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.9023 | 1.2735 | − 1.7439 | − 1.5252 |
| ACN | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.6988 | − 0.3690 | − 1.5571 | 0.3244 |
| LULU | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.8871 | − 1.1796 | − 0.2499 | 0.0056 |
| ADBE | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.8650 | − 1.1796 | − 1.6837 | 1.8182 |
| AMD | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.1653 | − 1.1796 | − 1.5016 | 0.9761 |
| AFL | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0.4089 | 0.1642 | − 0.1271 | 0.3415 |
| A | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.5763 | − 0.7617 | 0.0335 | − 1.3369 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| WY | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0.1772 | 0.0866 | 0.3793 | 0.5810 |
| WHR | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.3251 | 1.4469 | 1.4102 | 0.6982 |
| WYNN | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 1.2228 | − 0.6509 | 1.0537 | − 0.7490 |
| XEL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0.4151 | 0.6883 | 0.0685 | − 0.9215 |
| XYL | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0.0305 | − 0.5252 | 0.2913 | 0.2926 |
| YUM | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 1.7167 | − 1.1796 | − 0.0492 | 0.0821 |
| ZBRA | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | − 0.4738 | − 1.1796 | 0.8263 | − 0.8550 |
| ZBH | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0.1558 | − 0.8115 | 0.1791 | 0.6975 |
| ZION | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 2.5853 | 2.2292 | 2.0806 | − 1.1144 |

Notes: This table displays a portion of the factor loading matrix. The rows are labelled with stock ticker symbols, while the columns represent different factors. The 11 U.S. sector factors include Industrials (‘Ind’), Health Care (‘H Care’), Information Technology (‘IT’), Consumer Discretionary (‘Cons D’), Financials (‘Fina’), Materials (‘Mat’), Real Estate (‘R Est’), Utility (‘Util’), Communication Services (‘Comm’), Consumer Staples (‘Cons S’), and Energy (‘Energy’). Additionally, there are 4 fundamental factors: Book-to-Price Ratio (‘BP’), Dividend Yield (‘DY’), Size (‘Size’) (SMB type), and 3-Month Price Momentum (‘Ret_3M’).

yielding a 2.5% risk-free rate†. Monthly mean predictions (views) are simulated at varying investor skill levels (Subsection 3.2.3), while covariance matrices are estimated via the multifactor model (1); their interpretation, however, is subject to the investor’s subjective judgement. Portfolio performance is based on S&P Capital IQ historical close prices. Portfolios are launched on 01 March 2013 and rebalanced monthly through 02 October 2023 ($T = 128$ periods)‡, with rebalances on the first trading day of each month. For a pure efficiency assessment, fractional-share holdings are allowed, while frictions such as slippage, transaction costs, and taxes are ignored (as justified in Subsection 3.2.1).

### 3.2. *Methodology for portfolio-construction model evaluation*

This methodological formulation aims to (1) isolate portfolio-selection (PS) models for independent evaluation, addressing the joint-test problem; (2) provide a level playing field for comparison; and (3) assess the impact of investor *skill* and *subjectivity* on portfolio performance.

**3.2.1. Unified portfolio generation: ensuring a level playing field.** Suppose cognitively bounded investors have the following information – assuming normality, consistent with Paper (I) – for a universe of $n$ securities at time $t$:

$$\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{\mu}}_t, \boldsymbol{\Sigma}_t) \tag{2}$$

where $\widetilde{\boldsymbol{r}}_t$ is the $[n \times 1]$ vector of excess security returns (hereafter, ‘return’ denotes ‘excess return’ unless otherwise specified); $\mathbb{N}(\cdot)$ denotes the normal distribution; $\widehat{\boldsymbol{\mu}}_t$ is the $[n \times 1]$ mean vector; and $\boldsymbol{\Sigma}_t$ is the $[n \times n]$ covariance matrix.

Based on their interpretation of (2), investors form subjective views:

$$\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{q}}_t, \boldsymbol{\Omega}_t) \tag{3}$$

where $\widehat{\boldsymbol{q}}_t$ is the $[n \times 1]$ view *signal* vector, containing the investor’s forward-looking beliefs on security returns§; and $\boldsymbol{\Omega}_t$ is the $[n \times n]$ view *noise* matrix, representing the investor’s forward-looking beliefs on view uncertainties.

Investors construct portfolios based on their subjective views (3), rather than the raw information (2), and are assumed to have full discretion in forming these views. Rather than proposing normative rules, this paper considers nine subjectivity scenarios (Subsection 3.2.2) reflecting real-world investor behaviour. Portfolios for comparison are generated using a unified SAR-based procedure (Hypothesis 4.4 (Subjective Allocation Rule), Paper (I)):

(1) **Obtain the raw SAR portfolio**:

$$\boldsymbol{w}_t^* \propto \boldsymbol{\Omega}_t^{-1} \widehat{\boldsymbol{q}}_t \tag{4}$$

where $\boldsymbol{w}_t^*$ is the $[n \times 1]$ vector of optimal portfolio weights; and $\propto$ denotes direct proportionality.

(2) **Cap gross weights**: To ensure fair comparisons, portfolio gross weights (sum of absolute weights) are

† This approximates the historical average for the testing period. While different assumptions affect absolute portfolio performance, relative comparisons remain largely unaffected.
‡ Exhaustively testing all periods is impractical, but findings are robust to the choice of testing period.

§ To test SAR, the number of views is assumed equal to the number of securities.

Table 2. SAR-based simple and unified portfolio generation.

| | $\mathbf{1}^{\mathrm{T}}\lvert \boldsymbol{w}_t^* \rvert \leqslant 1$ | $\mathbf{1}^{\mathrm{T}}\lvert \boldsymbol{w}_t^* \rvert > 1$ |
|---|---|---|
| Portfolio | $\boldsymbol{w}_t^* \leftarrow \boldsymbol{w}_t^*$ | $\boldsymbol{w}_t^* \leftarrow \dfrac{\boldsymbol{w}_t^*}{\mathbf{1}^{\mathrm{T}}\lvert \boldsymbol{w}_t^* \rvert}$ |
| Treasury bond | $w_c \leftarrow 1 - \mathbf{1}^{\mathrm{T}}\boldsymbol{w}_t^*$ | |

Notes: Simple and unified portfolio construction procedure: Obtain the raw SAR portfolio $\boldsymbol{w}_t^* = \boldsymbol{\Omega}_t^{-1}\widehat{\boldsymbol{q}}_t$ given views $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{q}}_t, \boldsymbol{\Omega}_t)$. If the gross weight $\mathbf{1}^{\mathrm{T}}\lvert \boldsymbol{w}_t^* \rvert$ exceeds 100%, $\boldsymbol{w}_t^*$ is scaled down to normalise it to 100%; otherwise, $\boldsymbol{w}_t^*$ is kept as is and the cash surplus is invested into treasury bonds.

capped at 100% via homogeneous scaling (Table 2).† Portfolios exceeding 100% are scaled down; otherwise, weights remain unchanged and cash surplus is invested risk-free. Without efficiency-undermining constraints, all excess return reflects portfolio selection.

(3) **Fix monthly weights**: On rebalancing days, weights are calculated in shares and held fixed until the next rebalancing.

(4) **Compute portfolio value**: Portfolio value is updated, accounting for cash compounding at the risk-free rate.

This unified process ensures fair comparisons of PS models if the following principles are met: (1) **Identical information**: all models receive the same information Eq (2), subject to investors' subjective interpretation Eq (3); (2) **Identical constraints**: any constraints are applied uniformly across models; (3) **Minimal distortions**: constraints and frictions‡ are minimised to avoid information distortion and model-specific knock-on effects (e.g. MV's error amplification).

Since SAR is tested as a stand-alone technique, the evaluation provides general insights into the allocation principle beyond the ABL model (Cheung 2013) from which it is derived.

**3.2.2. *Comparative subjectivity testing* (CST): assessing the economic value of subjectivity.** As shown in Figure 1, on each rebalancing day $t$ (the first day of month $t$, $1 \leqslant t \leqslant T = 128$), previous-period estimates provide the *ex post* distribution $\mathbb{N}(\widehat{\boldsymbol{\mu}}_{t-1}, \boldsymbol{\Sigma}_{t-1})$, where $\widehat{\boldsymbol{\mu}}_{t-1}$ is the $[n \times 1]$ return mean vector ($n = 448$) and $\boldsymbol{\Sigma}_{t-1}$ the $[n \times n]$ covariance matrix via factor model (1). Subsequent-period estimates define the unobservable *ex ante* distribution $\mathbb{N}(\widehat{\boldsymbol{\mu}}_t, \boldsymbol{\Sigma}_t)$, which investors aim to predict for portfolio construction and rebalancing.

Assume the *ex ante* mean vector, $\widehat{\boldsymbol{\mu}}_t$, is partially predictable, depending on the investor's forecasting skill. Following Jensen (1972), *forecasting skill* or *information coefficient* (IC) is the correlation between the investor's mean forecasts, $\widehat{\boldsymbol{y}}_{t[n\times1]}$, and the realised (*ex ante*) means, $\widehat{\boldsymbol{\mu}}_t$. Whereas the *ex ante* covariance matrix, $\boldsymbol{\Sigma}_t$, is assumed unpredictable; investors rely on the latest risk update, $\boldsymbol{\Sigma}_{t-1}$, which they may accept or interpret subjectively. Investor *skill* and *subjectivity* thus generate the view scenarios described below.

Table 3 presents the nine behaviour-implied view scenarios with varying subjectivity, capturing diverse real-world investor behaviours compared in this paper.

(1) Portfolio P11, the *historical mean-variance (MV) portfolio*, assumes views $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{\mu}}_{t-1}, \boldsymbol{\Sigma}_{t-1})$, meaning the investor fully trusts historical mean and variance estimates as forecasts. Substituting these into the portfolio-generation procedure gives the unconstrained, but uniquely scaled (see Table 2) MV portfolio: $\boldsymbol{w}_t^* \propto \boldsymbol{\Sigma}_{t-1}^{-1}\widehat{\boldsymbol{\mu}}_{t-1}$. As this paper compares broadly different investor behaviours rather than optimisation nuances, P11 represents the family of historical MV portfolios, constrained or not, consistent with the fairness principles in Subsection 3.2.1.

(2) P21, the *historical mean-risk portfolio* ('risk' refers to marginal variances), assumes $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{\mu}}_{t-1}, \mathrm{diag}(\boldsymbol{\Sigma}_{t-1}))$, where $\mathrm{diag}(\boldsymbol{\Sigma}_{t-1})$ retains only diagonal elements, ignoring stock correlations. It suits investors who are correlation-agnostic or treat historical correlations as negligible, effectively assuming independent views. P21 weighs stocks by return-to-variance ratios: $\boldsymbol{w}_t^* \propto [\mathrm{diag}(\boldsymbol{\Sigma}_{t-1})]^{-1}\widehat{\boldsymbol{\mu}}_{t-1}$. Practical variants include volatility-managed portfolios using historical means (e.g. Moreira and Muir 2017).

(3) P31, the *historical mean-only portfolio*, assumes $\widetilde{r}_t \sim \mathbb{N}(\widehat{\boldsymbol{\mu}}_{t-1}, \mathbf{I}_n)$, where $\mathbf{I}_n$ is the $n$-dimensional identity matrix. It applies when investors do not use a risk model or ignore the covariance matrix $\boldsymbol{\Sigma}_{t-1}$. P31 weighs stocks by historical means: $\boldsymbol{w}_t^* \propto \widehat{\boldsymbol{\mu}}_{t-1}$, representing stock pickers' convictions based on past price momentum, regardless of risks or correlations.

By setting all covariances to 0, P21 and P31 effectively represent orthogonal-view scenarios, consistent with Assumption 4.2 (Correlation-Agnostic Orthogonal Views) of Paper (I) and reflecting human cognitive limits in specifying view correlations.

Suppose an investor with skill $\rho$ prefers their own forecasts, $\widehat{\boldsymbol{y}}_t$, over historical means, $\widehat{\boldsymbol{\mu}}_{t-1}$, as predictions (signals) for the next period.

(4) Portfolio P12, the *signal-variance portfolio*, assumes $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{y}}_t, \boldsymbol{\Sigma}_{t-1})$. Investors trust their own mean forecasts, $\widehat{\boldsymbol{y}}_t$, for the next period while relying on historical variance, $\boldsymbol{\Sigma}_{t-1}$. P12 weighs stocks as $\boldsymbol{w}_t^* \propto \boldsymbol{\Sigma}_{t-1}^{-1}\widehat{\boldsymbol{y}}_t$, featuring 'modern-sense MV optimisation' where historical means are replaced by investor-generated forecasts.

(5) P22, the *signal-risk portfolio*, assumes $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{y}}_t, \mathrm{diag}(\boldsymbol{\Sigma}_{t-1}))$, ignoring correlations. Like P21, it suits investors who are correlation-agnostic or sceptical of historical correlations. P22 weighs stocks by unit-variance signals: $\boldsymbol{w}_t^* \propto [\mathrm{diag}(\boldsymbol{\Sigma}_{t-1})]^{-1}\widehat{\boldsymbol{y}}_t$. Practical variants include volatility-managed portfolios using forward-looking signals.

(6) P32, the *signal-only portfolio*, assumes $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{y}}_t, \mathbf{I}_n)$, relying solely on investor forecasts. Like P31, it applies when no risk model is used or trusted.

† Homogeneous scaling is chosen over the fully-invested constraint ($\boldsymbol{w}^{\mathrm{T}}\mathbf{1}_n = 1$, where $\mathbf{1}_n$ is the $[n \times 1]$ column vector of ones) to minimise information distortion and preserve (4).

‡ Factors like slippage, transaction costs, taxes, and lot-size constraints affect performance in specific strategies but may distort relative PS model comparisons, introducing joint-test issues.

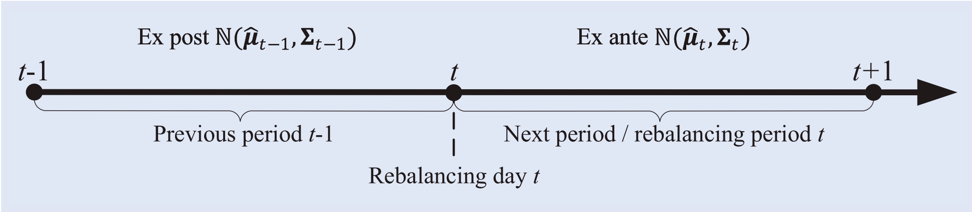


Figure 1. Ex post vs Ex Ante estimates on a rebalancing day.

Table 3. View scenarios reflecting varying levels of investor subjectivity.

| | Signal | | |
|---|---|---|---|
| Noise | Historical mean $\widehat{\boldsymbol{q}}_t = \widehat{\boldsymbol{\mu}}_{t-1}$(*ex post*) | Skill-$\rho$ signal $\widehat{\boldsymbol{q}}_t = \widehat{\boldsymbol{y}}_t$(*ex ante* view) | Skill-$\rho$ directional view (DV) $\widehat{\boldsymbol{q}}_t = \text{sign}(\widehat{\boldsymbol{y}}_t)$ |
| Historical variance $\boldsymbol{\Omega}_t = \boldsymbol{\Sigma}_{t-1}$(*ex post* est.) | Hist M-V P11 | Signal-V P12 | DV-V P13 |
| Historical risks $\boldsymbol{\Omega}_t = \text{diag}(\boldsymbol{\Sigma}_{t-1})$ | Hist M-risk P21 | Signal-risk P22 | DV-risk P23 |
| No risk information $\boldsymbol{\Omega}_t = \mathbf{I}_n$ | Hist M only P31 | Signal only P32 | Qualitative P33 |

Notes: Denoting investor views as $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\widehat{\boldsymbol{q}}_t, \boldsymbol{\Omega}_t)$, this table presents combinations of view signals ($\widehat{\boldsymbol{q}}_t$) and noise ($\boldsymbol{\Omega}_t$). The SAR maps these view scenarios into 9 portfolio strategies (P11-P33) for out-of-sample performance comparison, illustrating the impact of view subjectivity on portfolio's performance.

P32 weighs stocks by signals: $\boldsymbol{w}_t^* \propto \widehat{\boldsymbol{y}}_t$, featuring long/short, conviction-weighted stock-picking strategies based on forecasts, irrespective of risks or correlations.

Note by setting all covariances to 0, P22 and P32 effectively represent orthogonal-view scenarios.

Suppose an investor receiving skill-$\rho$ forecasts only trusts the directional views (DV) $\text{sign}(\widehat{\boldsymbol{y}}_t)$, where $\text{sign}(\cdot)$ leaves 0 unchanged, converts positives to $+1$ and negatives to $-1$. Even with quantitative signals $\widehat{\boldsymbol{y}}_t$, the investor relies only on direction, ignoring magnitude. DVs align with Assumption 4.1 (Strength-Agnostic Directional Views) in Paper 1, and with analyst ratings in practice, which typically follow an *ordinal* scale (Buy $>$ Hold $>$ Sell).

(7) P13, the *DV-variance portfolio*, assumes $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\text{sign}(\widehat{\boldsymbol{y}}_t), \boldsymbol{\Sigma}_{t-1})$ and weighs stocks as $\boldsymbol{w}_t^* \propto \boldsymbol{\Sigma}_{t-1}^{-1}\text{sign}(\widehat{\boldsymbol{y}}_t)$. It resembles a two-tiered quantamental approach, where analysts recommend stocks, and portfolio managers optimise the portfolio using equal-magnitude long/short signals against a risk model to control risk.

(8) P23, the *DV-risk portfolio*, assumes $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\text{sign}(\widehat{\boldsymbol{y}}_t), \text{diag}(\boldsymbol{\Sigma}_{t-1}))$, ignoring correlations like P21. It allocates as $\boldsymbol{w}_t^* \propto [\text{diag}(\boldsymbol{\Sigma}_{t-1})]^{-1}\text{sign}(\widehat{\boldsymbol{y}}_t)$, representing a volatility-managed stock-picking strategy.

(9) P33, the *qualitative investing portfolio*, assumes $\widetilde{\boldsymbol{r}}_t \sim \mathbb{N}(\text{sign}(\widehat{\boldsymbol{y}}_t), \mathbf{I}_n)$ and weighs stocks equally: $\boldsymbol{w}_t^* \propto \text{sign}(\widehat{\boldsymbol{y}}_t)$. This long/short equal-weighted ('LS 1/N') strategy captures a fundamental stock-picking approach.

Again, P23 and P33 represent orthogonal-view scenarios.

*Enabling evaluation of subjectivity as a portfolio performance factor.* Under SAR, view scenarios capture investor attitudes towards mean prediction and variance estimation. On the *signal* side: (1) $\widehat{\boldsymbol{q}}_t = \widehat{\boldsymbol{\mu}}_{t-1}$ uses historical means; (2) $\widehat{\boldsymbol{q}}_t = \widehat{\boldsymbol{y}}_t$ reflects forecasting skill; (3) $\widehat{\boldsymbol{q}}_t = \text{sign}(\widehat{\boldsymbol{y}}_t)$ ignores signal magnitude, consistent with Assumption 4.1 (Strength-Agnostic Directional Views) of Paper (I). On the *noise* side: (1) $\boldsymbol{\Omega}_t = \boldsymbol{\Sigma}_{t-1}$ assumes stable variance-covariance; (2) $\boldsymbol{\Omega}_t = \text{diag}(\boldsymbol{\Sigma}_{t-1})$ assumes stable variances but noisy or irrelevant correlations; (3) $\boldsymbol{\Omega}_t = \mathbf{I}_n$ disregards historical variances entirely. The latter two treat views as orthogonal, allowing the consequence of correlation agnosticism to be tested, in line with Assumption 4.2 (Correlation-Agnostic Orthogonal Views) of Paper (I).

Under SAR, subjective interpretations of the same information yield distinct real-world PS behaviours. This unifying feature enables a **Comparative Subjectivity Testing** (CST) framework†: *attributing differences across PS models solely to variations in subjectivity and allowing empirical assessment of its economic value*. Building on CST, portfolios P12, P22, P32, P13, P23, and P33 – differing only in how they process the same information, $\mathbb{N}(\widehat{\boldsymbol{y}}_t, \boldsymbol{\Sigma}_{t-1})$ – reveal the impact of investor subjectivity on portfolio performance. In particular, comparing P22, P32, P23, and P33 – corresponding to different treatments of the covariance matrix – enables evaluation of portfolio massage (Definition 4.1, Paper (I)) and orthogonal representation of correlated views (Lemma 4.5).

### 3.2.3. *Skill-controlled view simulations* (SVS): isolating PS model evaluation.

Suppose an investor's prediction skill is uniform across all forecasts. For each month $t$, their predicted

† Just as comparative statics analyses equilibrium changes when an exogenous parameter varies, CST compares portfolio outcomes across subjective interpretations of the same information.

views can be generated via Monte Carlo simulation as:

$$\widetilde{\mathbf{y}}_t|_{\widetilde{\boldsymbol{\mu}}=\widehat{\mu}_t} = \rho\widehat{\boldsymbol{\mu}}_t + \sqrt{1-\rho^2}\widetilde{\boldsymbol{\mu}}_t^{\perp} \; (1 \leqslant t \leqslant T) \tag{5}$$

where $\widetilde{\mathbf{y}}_t$ is the month-$t$ return outlook; $\widetilde{\boldsymbol{\mu}}$ is the monthly stock return vector; $\widehat{\boldsymbol{\mu}}_t$ is the *ex ante* month-$t$ stock return realisation vector (see Figure 1); and $\widetilde{\boldsymbol{\mu}}_t^{\perp}$ is an $[n \times 1]$ innovation vector independently and identically distributed (i.i.d.) as $\widetilde{\boldsymbol{\mu}}$.

For each stock $i$ $(1 \leqslant i \leqslant n)$, the simulated series $\{\widehat{y}_{it}$ $(1 \leqslant t \leqslant T)\}$ is $\rho$-correlated with the *ex ante* returns $\{\widehat{\mu}_{it}$ $(1 \leqslant t \leqslant T)\}$, sharing the same variance and representing skill-based predictions. This approach, similar to Allen *et al.* (2019) but explicitly controls for skill, generates means without relying on any alpha model, and avoids the joint-test issue, enabling a clean evaluation of the PS model.

*Enabling evaluation of skill as a portfolio performance factor.* By controlling for investor prediction skill, the sensitivity of the PS model's performance can be assessed to validate Lemma 4.2 (SAR Alpha Generation with Orthogonal Views, Paper I), simultaneously addressing the traditional *joint-test* issue by disentangling skill from model quality.

Several skill levels are considered: mildly negative skill ($\rho = -0.01$), almost zero skill ($\rho = 0.001$), low skill ($\rho = 0.03$), medium skill ($\rho = 0.07$), high skill ($\rho = 0.10$), and extra high ($\rho = 0.15$). For each level, $n$ stock-view trajectories $\{\widehat{\mathbf{y}}_t$ $(1 \leqslant t \leqslant T)\}$ are simulated; portfolios P12, P22, P32, P13, P23, and P33 are generated, rebalanced monthly, and their daily OOS performances recorded. The procedure is repeated 100 times per skill level to produce statistics.

**3.2.4. Jensen's Information Ratio: A Principled Measure of Active-Performance Efficiency.** Building on Fama (1972), who uses Jensen's alpha to measure *selectivity* skill premium, Jensen's information ratio (JIR) quantifies PS models' *skill-premium efficiency*, i.e. how effectively prediction skill translates into *active* portfolio gains. Since Jensen's alpha measures excess return over the CAPM risk premium, JIR is preferred over the standard information ratio (IR), which can misrepresent extreme-$\beta$ strategies.

### *3.3. Results and analysis*

Table 4 shows medium-skill ($\rho = 0.07$) portfolio P12 (Signal-Variance) OOS performances across 100 simulated view histories, with each simulation in a row. High average-to-standard-error ratios indicate stable, consistent results, confirming that 100 simulations suffice for convergence.

The P12 'Average' row is stacked with those of P22, P32, P13, P23, and P33 at the same skill level to form the 'Medium skill ($\rho = 0.07$)' panel in Table 5. Stacking this panel alongside its counterparts from other skill levels ($\rho = -0.01, 0.001, 0.03, 0.10, 0.15$) produces a complete table that reveals how view skill affects OOS portfolio performance and how these portfolios compare with historical mean-based portfolios P11, P21, P31, and the Market.

*The market.* To avoid the universe-shift effects of index-rebalancing, the market-cap-weighted portfolio P0 (based on the fixed universe) serves as the benchmark instead of the SPX. Over the testing period, P0 achieves an annualised return of 10.13%, risk of 17.55%, and SR of 0.43.

**3.3.1. Performance of historical mean-based portfolios.** *The historical mean-based portfolios.* P11 (historical MV portfolio) delivers an annualised return of 1.20% with 3.35% risk, below the 2.5% risk-free rate (SR = −0.39). P21 (historical mean–risk) and P31 (historical mean-only) lose money, with SRs of −0.48 and −0.40, respectively. P11 appears more conservative, exhibiting much lower OOS risk than P21 and P31. This is because the raw SAR portfolio, based on the full historical covariance matrix, holds large long and short positions. Gross-weight normalisation then scales these down to near-trivial levels, leaving much of the capital earning the risk-free rate. Hence, although it earns a positive return, its SR is negative and similar to that of the other two.

*The 'historical mean skill'.* Monthly return means exhibit weak predictive power: the average 1-month lagged autocorrelation is −0.0855, indicating a dominant reversal effect over momentum. This poor 'historical mean skill' explains the negative OOS SRs of P11, P21, and P31.

**3.3.2. Cross-sectional observations based on *Skill*.** *The mildly negative skill ($\rho = -0.01$) portfolios.* Despite slightly negative skill, all six view portfolios deliver positive returns. Their long/short structures hold substantial cash earning the risk-free rate, yielding about 2.5%. P12 and P13, which use the full historical covariance matrix, have lower OOS risk but underperform the risk-free rate, resulting in negative SRs. They also lag behind their risk-only (P22, P23) and risk-unaware (P32, P33) counterparts. Their negative IRs and JIRs illustrate how poor skill erodes active performance. However, compared with the above historical mean-based portfolios (skill of −0.0855), the $\rho = -0.01$ portfolios outperform across all risk-adjusted metrics – showing that reliance on historical means can be even more detrimental to OOS performance than mildly negative forecasting skill.

*The almost zero skill ($\rho = 0.001$) portfolios.* All $\rho = 0.001$ portfolios *stochastically dominate*† their $\rho = -0.01$ counterparts, exhibiting higher OOS returns and lower risks. They begin to exhibit positive SRs and JIRs, though outperformance over the benchmark remains marginal. This aligns with Lemma 4.2 of Paper (I), which links SAR outperformance to view skill: negative skills cause underperformance, zero skills add no value, and positive skills generate alpha.

*The positive skill ($\rho = 0.03, 0.07, 0.10,$ and $0.15$) portfolios.* All positively skilled portfolios outperform the risk-free rate, yielding positive SRs. As skill rises, OOS mean returns increase monotonically, with only modest risk growth, producing a clear pattern of SR improvement. This supports Lemma 4.2 of Paper (I), which states that SAR value is proportional to view skill. Given that Table 5 portfolios are based on monthly return predictions, which may correlate weakly with the market's returns, their betas remain relatively low.

*Skill versus performance.* Figure 2 compares all view portfolios via JIR, representing skill-premium efficiency, across skill levels. Key observations: (1) Within each category, JIRs scale with view skill. (2) Signal view portfolios (P12, P22, P32) outperform their DV counterparts (P13, P23, P33),

† Assuming normality, *first-order stochastic dominance* is based on mean-variance comparisons, with dominance implying higher mean return and lower variance.

Table 4. Medium skill ($\rho = 0.07$) portfolio P12 OOS performance across simulations.

| | Mean | Risk | SR | Beta | Alpha | Active Risk | IR | Jensen Alpha | Jensen IR |
|---|---|---|---|---|---|---|---|---|---|
| Sim 1 | 6.96% | 1.28% | 3.48 | 0.00 | – 3.17% | 17.57% | – 0.18 | 4.45% | 0.25 |
| Sim 2 | 6.63% | 1.22% | 3.38 | – 0.00 | – 3.51% | 17.63% | – 0.20 | 4.14% | 0.24 |
| Sim 3 | 6.27% | 1.23% | 3.06 | 0.00 | – 3.86% | 17.57% | – 0.22 | 3.77% | 0.21 |
| Sim 4 | 6.63% | 1.23% | 3.35 | 0.00 | – 3.50% | 17.52% | – 0.20 | 4.10% | 0.23 |
| Sim 5 | 6.30% | 1.25% | 3.03 | 0.00 | – 3.83% | 17.53% | – 0.22 | 3.77% | 0.22 |
| … | … | … | … | … | … | … | … | … | … |
| Sim 96 | 6.57% | 1.38% | 2.95 | 0.00 | – 3.56% | 17.57% | – 0.20 | 4.06% | 0.23 |
| Sim 97 | 6.66% | 1.22% | 3.40 | 0.00 | – 3.47% | 17.53% | – 0.20 | 4.14% | 0.24 |
| Sim 98 | 6.63% | 1.25% | 3.31 | 0.00 | – 3.50% | 17.56% | – 0.20 | 4.12% | 0.23 |
| Sim 99 | 7.28% | 1.29% | 3.71 | 0.00 | – 2.85% | 17.58% | – 0.16 | 4.78% | 0.27 |
| Sim 100 | 6.99% | 1.27% | 3.52 | 0.00 | – 3.15% | 17.56% | – 0.18 | 4.47% | 0.25 |
| Average | 6.58% | 1.25% | 3.27 | 0.00 | – 3.55% | 17.55% | – 0.20 | 4.06% | 0.23 |
| Std Err | 0.034% | 0.004% | 0.027 | 0.000 | 0.034% | 0.004% | 0.002 | 0.034% | 0.002 |

suggesting subjective adjustments to positive-skilled forecasts can lead to a modest attenuation of performance.

Observation 3.1 (Portfolio Performance vs Mean Skill) *SAR generates a Jensen's information ratio (JIR) in proportion to mean-prediction skill. Skilled quantitative signals should be implemented at full scale; however, if full quantitative signals are unattainable, even directional views retain the bulk of the value proportional to skill.*

Remark This is consistent with Lemma 4.2 of Paper (I): SAR generates an IR proportional to view skill. Even when investors cannot fully quantify view strength, SAR still adds value proportionally to skill, supporting Assumption 4.1 (Strength-Agnostic Directional Views). This observation resonates with signal-processing theory: for Gaussian signals, a 1-bit sign quantiser (i.e. a DV) deterministically retains $\sqrt{2/\pi} \approx 80\%$ of the original predictive skill (linear correlation) (Bussgang theorem), illustrating why DVs can still preserve most of the value of skilled forecasts.

Forthcoming research (Cheung, 2026b) examines Lemma 4.2's claim regarding the effect of the betting scale on the IR.

#### 3.3.3. Cross-sectional observations based on *Subjectivity*.

*Subjectivity versus performance.* Figure 2 shows that risk-agnostic portfolios P32 (signal only portfolio) and P33 (equal-weighted qualitative investing) consistently outperform their marginal-risk-aware (P22, P23) and fully risk-aware (P12, P13) counterparts. Table 5 confirms that, across all positive skill levels, P32 $\succ$ P22 $\succ$ P12 and P33 $\succ$ P23 $\succ$ P13 (where '$\succ$' denotes 'stochastic dominance') in terms of OOS Jensen's alpha and active risk. For example, at medium skill ($\rho = 0.07$), P32 achieves higher Jensen's alpha (10.70%) and lower active risk (15.52%) than P22 (9.10%, 15.70%) and P12 (4.06%, 17.55%). Since these portfolios share the same means, differences arise solely from subjective treatment of the covariance matrix, highlighting that *investor subjectivity can enhance performance*.

Note that all top-performing portfolios treat views as orthogonal – ignoring estimated correlations. This challenges the perceived importance of covariance estimation accuracy.

Observation 3.2 (Portfolio Performance vs Covariance Subjectivity) *With positive mean-prediction skill, risk-agnostic portfolios can dominate their risk-aware counterparts, including MV portfolios. Appropriate subjective treatments of error-prone covariance estimates can significantly enhance performance.*

Remark This observation highlights that subjective adjustments to the covariance matrix can enhance portfolio performance, including (1) *portfolio massage* (adjusting marginal variances or view noises) as given in Definition 4.1 of Paper (I), and (2) *treating views as independent* (i.e. ignoring correlations), as formalised in Lemma 4.5. Such treatments generalise covariance matrix *shrinkage*, which may be implemented through optimisation (e.g. Ledoit and Wolf 2003) or through the more flexble *portfolio massage* mechanism developed in Paper (I).

Since P12 is a modern-sense MV portfolio using investor forecasts, whereas P33 is a long/short equal-weighted ('LS 1/N') portfolio, Observation 3.2 aligns with DeMiguel *et al.* (2009), Fischer and Gallmeyer (2016), and Yuan and Zhou (2024), who find that naïve 1/N strategies can outperform MV portfolios.

The interpretation of Observation 3.2 is twofold. First, subjective fuzzifications of the covariance matrix can mitigate estimation errors. The MV optimiser's sensitivity to input errors often makes the required MV estimation accuracy unrealistic. Losses from estimation errors can outweigh gains from strict covariance consideration, whereas fuzzy approaches – like P32 (signal-only) and P33 (LS 1/N) – may better translate investor skill into performance. This challenges the traditional view that performance depends primarily on estimation accuracy. Cheung (2024b) examines the thresholds at which subjective interventions become crucial for achieving performance breakthroughs. Second, variance, while affecting investor psychology, does not always reflect true profit-and-loss (P&L) risks. Variance-based optimisation assumes regular rebalancing, but patient investors can wait for favourable conditions. Strict reliance on variance estimates can lead to premature or delayed rebalancing, or missed opportunities, causing underperformance.

#### 3.3.4. Unification.

Additionally, the tests highlight SAR's role as a unified portfolio generator:

Observation 3.3 (The Unifying Power of SAR) *By allowing information-processing subjectivity, SAR unifies a spectrum*

Table 5. Performance comparison of different view portfolios at varying skill levels.

| | Mean | Risk | SR | Beta | Alpha | Active Risk | IR | Jensen Alpha | Jensen IR |
|---|---|---|---|---|---|---|---|---|---|
| P0-MarketCap Weighted | 10.13% | 17.55% | 0.43 | 1.00 | 0.00% | 0.00% | NaN | 0.00% | Inf |
| Hist mean skill (rho = −0.0855) | | | | | | | | | |
| P11-Hist M-V | 1.20% | 3.35% | −0.39 | −0.03 | −8.93% | 18.41% | −0.49 | −1.05% | −0.06 |
| P21-Hist M-Risk | −3.95% | 13.35% | −0.48 | −0.17 | −14.08% | 24.34% | −0.58 | −5.13% | −0.21 |
| P31-Hist M Only | −3.24% | 14.36% | −0.40 | −0.19 | −13.37% | 25.13% | −0.53 | −4.29% | −0.17 |
| Negative skill (rho = −0.01) | | | | | | | | | |
| P12-Signal (rho = −0.01)-V | 2.35% | 1.24% | −0.12 | 0.00 | −7.78% | 17.54% | −0.44 | −0.17% | −0.01 |
| P22-Signal (rho = −0.01)-Risk | 3.49% | 3.32% | 0.30 | 0.16 | −6.64% | 14.90% | −0.45 | −0.21% | −0.01 |
| P32-Signal (rho = −0.01) Only | 3.67% | 3.83% | 0.31 | 0.17 | −6.46% | 14.75% | −0.44 | −0.13% | −0.01 |
| P13-DV (rho = −0.01)-V | 2.39% | 0.93% | −0.11 | 0.00 | −7.74% | 17.57% | −0.44 | −0.11% | −0.01 |
| P23-DV (rho = −0.01)-Risk | 3.09% | 2.22% | 0.27 | 0.10 | −7.04% | 15.87% | −0.44 | −0.16% | −0.01 |
| P33-Qualitative (rho = −0.01) | 3.17% | 2.38% | 0.28 | 0.11 | −6.96% | 15.75% | −0.44 | −0.14% | −0.01 |
| Almost zero skill (rho = 0.001) | | | | | | | | | |
| P12-Signal (rho = 0.001)-V | 2.96% | 1.23% | 0.37 | 0.00 | −7.18% | 17.54% | −0.41 | 0.43% | 0.02 |
| P22-Signal (rho = 0.001)-Risk | 4.74% | 3.27% | 0.69 | 0.15 | −5.40% | 14.97% | −0.36 | 1.06% | 0.07 |
| P32-Signal (rho = 0.001) Only | 5.16% | 3.81% | 0.70 | 0.17 | −4.97% | 14.81% | −0.34 | 1.39% | 0.09 |
| P13-DV (rho = 0.001)-V | 2.74% | 0.94% | 0.25 | 0.00 | −7.39% | 17.57% | −0.42 | 0.24% | 0.01 |
| P23-DV (rho = 0.001)-Risk | 3.82% | 2.18% | 0.61 | 0.10 | −6.31% | 15.92% | −0.40 | 0.59% | 0.04 |
| P33-Qualitative (rho = 0.001) | 4.01% | 2.37% | 0.64 | 0.10 | −6.12% | 15.79% | −0.39 | 0.72% | 0.05 |
| Low skill (rho = 0.03) | | | | | | | | | |
| P12-Signal (rho = 0.03)-V | 4.47% | 1.23% | 1.61 | 0.00 | −5.66% | 17.54% | −0.32 | 1.95% | 0.11 |
| P22-Signal (rho = 0.03)-Risk | 7.94% | 3.21% | 1.70 | 0.14 | −2.19% | 15.26% | −0.14 | 4.38% | 0.29 |
| P32-Signal (rho = 0.03) Only | 8.88% | 3.78% | 1.69 | 0.15 | −1.25% | 15.08% | −0.08 | 5.21% | 0.35 |
| P13-DV (rho = 0.03)-V | 3.62% | 0.94% | 1.20 | 0.00 | −6.51% | 17.57% | −0.37 | 1.12% | 0.06 |
| P23-DV (rho = 0.03)-Risk | 5.76% | 2.14% | 1.53 | 0.09 | −4.37% | 16.09% | −0.27 | 2.60% | 0.16 |
| P33-Qualitative (rho = 0.03) | 6.21% | 2.34% | 1.59 | 0.10 | −3.93% | 15.95% | −0.25 | 2.97% | 0.19 |
| Medium skill (rho = 0.07) | | | | | | | | | |
| P12-Signal (rho = 0.07)-V | 6.58% | 1.25% | 3.27 | 0.00 | −3.55% | 17.55% | −0.20 | 4.06% | 0.23 |
| P22-Signal (rho = 0.07)-Risk | 12.50% | 3.38% | 2.96 | 0.12 | 2.36% | 15.70% | 0.15 | 9.10% | 0.58 |
| P32-Signal (rho = 0.07) Only | 14.23% | 4.01% | 2.93 | 0.13 | 4.09% | 15.52% | 0.26 | 10.70% | 0.69 |
| P13-DV (rho = 0.07)-V | 4.87% | 0.95% | 2.50 | −0.00 | −5.26% | 17.57% | −0.30 | 2.37% | 0.13 |
| P23-DV (rho = 0.07)-Risk | 8.53% | 2.23% | 2.71 | 0.07 | −1.60% | 16.34% | −0.10 | 5.47% | 0.33 |
| P33-Qualitative (rho = 0.07) | 9.32% | 2.46% | 2.78 | 0.08 | −0.82% | 16.20% | −0.05 | 6.18% | 0.38 |
| High skill (rho = 0.10) | | | | | | | | | |
| P12-Signal (rho = 0.1)-V | 8.11% | 1.28% | 4.39 | 0.00 | −2.02% | 17.56% | −0.11 | 5.60% | 0.32 |
| P22-Signal (rho = 0.1)-Risk | 15.70% | 3.68% | 3.59 | 0.10 | 5.57% | 16.07% | 0.35 | 12.42% | 0.77 |
| P32-Signal (rho = 0.1) Only | 17.90% | 4.35% | 3.55 | 0.12 | 7.77% | 15.90% | 0.49 | 14.48% | 0.91 |
| P13-DV (rho = 0.1)-V | 5.75% | 0.96% | 3.39 | −0.00 | −4.38% | 17.58% | −0.25 | 3.26% | 0.19 |
| P23-DV (rho = 0.1)-Risk | 10.49% | 2.41% | 3.33 | 0.06 | 0.36% | 16.55% | 0.02 | 7.50% | 0.45 |
| P33-Qualitative (rho = 0.1) | 11.53% | 2.66% | 3.41 | 0.07 | 1.40% | 16.41% | 0.09 | 8.46% | 0.52 |
| Extra high skill (rho = 0.15) | | | | | | | | | |
| P12-Signal (rho = 0.15)-V | 10.68% | 1.35% | 6.06 | 0.00 | 0.55% | 17.56% | 0.03 | 8.17% | 0.47 |
| P22-Signal (rho = 0.15)-Risk | 21.05% | 4.45% | 4.18 | 0.08 | 10.92% | 16.70% | 0.65 | 17.95% | 1.07 |
| P32-Signal (rho = 0.15) Only | 24.11% | 5.15% | 4.20 | 0.10 | 13.98% | 16.55% | 0.84 | 20.86% | 1.26 |
| P13-DV (rho = 0.15)-V | 7.30% | 0.99% | 4.84 | −0.00 | −2.84% | 17.58% | −0.16 | 4.80% | 0.27 |
| P23-DV (rho = 0.15)-Risk | 13.77% | 2.87% | 3.94 | 0.05 | 3.64% | 16.88% | 0.22 | 10.88% | 0.64 |
| P33-Qualitative (rho = 0.15) | 15.21% | 3.16% | 4.03 | 0.06 | 5.08% | 16.74% | 0.30 | 12.24% | 0.73 |

Notes: This table compares the OOS performances of the Market and the historical MV portfolios with view portfolios across different skill levels.

*of investor behaviours, ranging from MV optimisation to qualitative investing (e.g. long/short 1/N strategies).*

REMARK SAR's ability to naturally align with diverse real-world investor allocation behaviours stems from the crucial shift from a traditional 'rocket-science' mindset to one of 'portfolio discovery': treating mean and variance estimates *subjectively* rather than *objectively*. It is this unifying property of SAR that enables the CST methodology.

## 4. Testing MTE via FM: how well FM tilts track factors?

This section validates the MTE principle (Proposition 4.1, Paper I) by assessing how effectively MTE-based factor

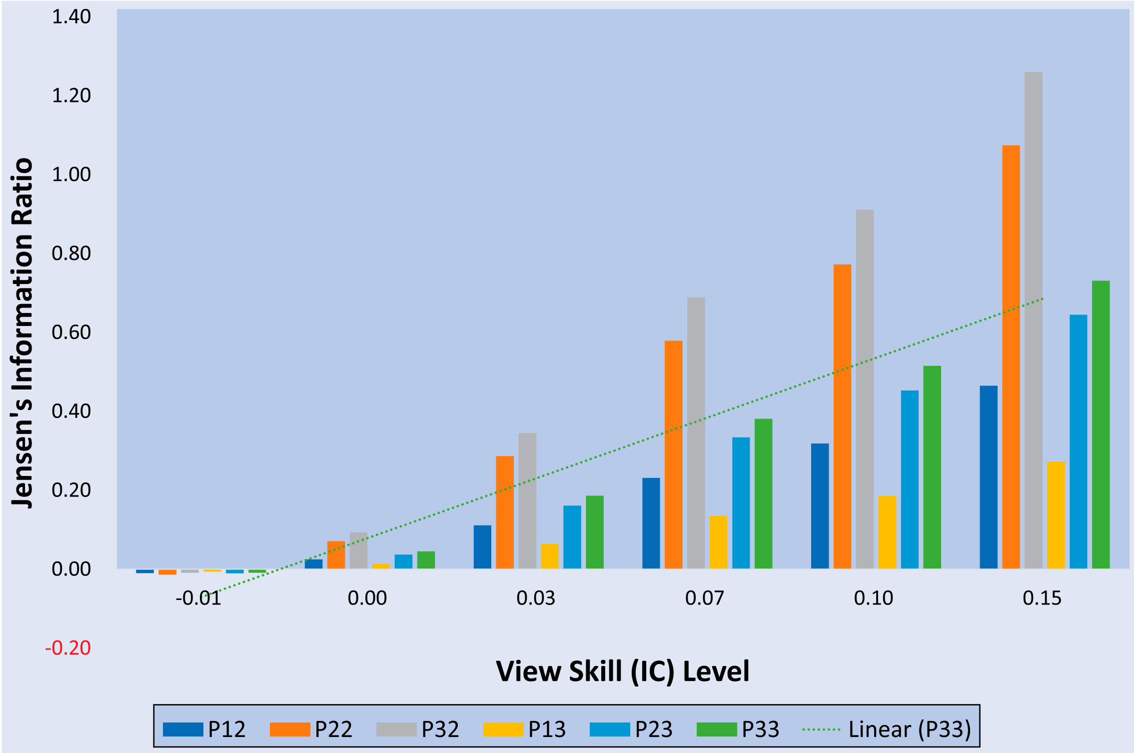


Figure 2. View portfolios' Jensen's information ratios (JIR) vs view skills (IC).
This figure compares view portfolio JIRs across different skill levels. Key observations are: (1) Signal view portfolios P12 (*signal-variance portfolio*), P22 (*signal-risk portfolio*) and P32 (*signal-only portfolio*) outperform their respective directional view (DV) counterparts P13 (*DV-variance portfolio*), P23 (*DV-risk portfolio*) and P33 (*qualitative investing*). (2) The risk-agnostic portfolios P32 and P33 consistently outperform their marginal riskaware counterparts P22 and P23, which further dominate their fully risk-aware counterparts P12 and P13.

view portfolios (*FM tilts*) mimic their target factors, using analytical exposure and OOS correlation.

### *4.1. Setup*

The analysis focuses on the 11 sectoral and 4 fundamental factors in the multifactor risk model (Table 1). The testing procedure is:

(1) Express a bullish view on a factor (e.g. Industrials or Size).
(2) Generate the latest portfolio (i.e. 01 November 2023) using the FM technique $\mathbf{M}_{\text{FM}}^{\text{T}}$ based on the October 2023 risk model, and examine its exposure to the target factor.
(3) Starting 01 Mar 2013, apply the factor view monthly, generate portfolios using the FM technique based on the preceding month's risk model, and calculate OOS daily returns through to 02 Oct 2023 (128 months).
(4) Compute the correlation between each OOS FM tilt return series and all 15 factor return series.
(5) Repeat for all 15 factors and assemble an OOS correlation matrix to evaluate how effectively FM portfolios mimic their target factors.

Since FM is tested as a stand-alone technique here, the results provide general insights into the MTE-based mimicking principle beyond the ABL model.

### *4.2. Results and analysis*

Figure 3 illustrates the behaviour of the Industrials FM tilt, a *sectoral* factor with binary stock exposures (Table 1). The left panel visualises its exposures to all 15 factors, where the *target exposure* to Industrials (orange) clearly dominates the others (blue), indicating the tilt is primarily sensitive to its target factor. Because stock-sector betas are binary in the multifactor risk model (Table 1), the tilt's sector exposures mirror its aggregate sector weights. Hence, the left panel exhibits a clear allocation bias towards the target sector, consistent with Cheung (2013). The right panel shows how closely the FM tilt (blue) tracks its target factor (orange) OOS over 128 months, yielding a 97% OOS return correlation with the target Industrials factor (Table 6).

Figure 4 illustrates the behaviour of the Size FM tilt, a *fundamental* factor with stock exposures represented by normalised stock characteristics (Table 1). The left panel exhibits a dominant bias towards Size, while the right panel shows the tilt closely tracking its target factor OOS over 128 months, with an 82% return correlation (Table 6).

Appendix B presents similar results for the other 13 factors.

Table 6 presents the [15 × 15] OOS correlation matrix between FM tilts and factors. Diagonal elements give each tilt's *target correlation* with its corresponding factor, while off-diagonals contain *non-target correlations*. For instance, the Industrials tilt correlates 97% with its factor, and the Size tilt 82%. Overall, FM tilts perform strongly OOS – sectoral

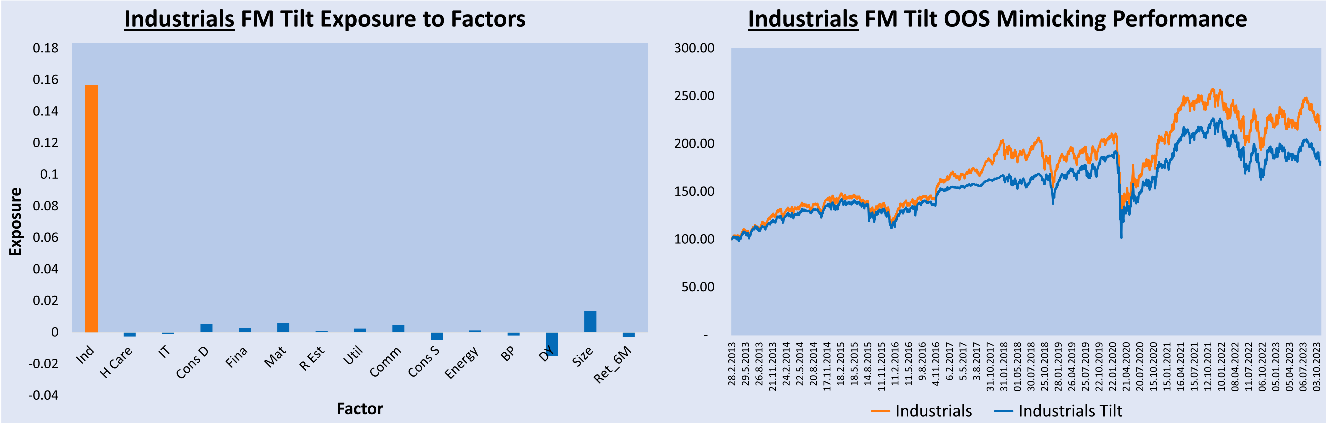


Figure 3. Industrials FM tilt exposures and OOS mimicking performance.

Table 6. OOS correlation matrix between factors and their mimicking tilts.

| | Ind | H Care | IT | Cons D | Fina | Mat | R Estate | Util | Comm | Cons S | Energy | BP | DY | Size | Ret_3M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Industrials-Tilt | 0.97 | 0.82 | 0.84 | 0.89 | 0.91 | 0.89 | 0.76 | 0.61 | 0.79 | 0.78 | 0.65 | 0.35 | 0.01 | 0.22 | -0.14 |
| Health Care-Tilt | 0.84 | 0.97 | 0.80 | 0.80 | 0.86 | 0.80 | 0.75 | 0.66 | 0.75 | 0.78 | 0.56 | 0.19 | -0.05 | 0.15 | -0.10 |
| Info Tech-Tilt | 0.88 | 0.82 | 0.97 | 0.86 | 0.87 | 0.83 | 0.73 | 0.57 | 0.82 | 0.72 | 0.62 | 0.22 | -0.17 | 0.14 | -0.09 |
| Consumer Disc-Tilt | 0.92 | 0.81 | 0.85 | 0.97 | 0.89 | 0.86 | 0.78 | 0.59 | 0.82 | 0.77 | 0.63 | 0.32 | -0.07 | 0.25 | -0.14 |
| Financials-Tilt | 0.91 | 0.83 | 0.83 | 0.86 | 0.98 | 0.85 | 0.76 | 0.62 | 0.79 | 0.78 | 0.62 | 0.31 | -0.02 | 0.16 | -0.15 |
| Materials-Tilt | 0.92 | 0.81 | 0.83 | 0.86 | 0.88 | 0.96 | 0.74 | 0.59 | 0.77 | 0.76 | 0.67 | 0.35 | -0.01 | 0.22 | -0.13 |
| Real Estate-Tilt | 0.73 | 0.73 | 0.69 | 0.74 | 0.75 | 0.69 | 0.98 | 0.77 | 0.67 | 0.73 | 0.48 | 0.11 | -0.05 | 0.13 | -0.17 |
| Utilities-Tilt | 0.58 | 0.62 | 0.51 | 0.52 | 0.59 | 0.53 | 0.74 | 0.95 | 0.51 | 0.69 | 0.34 | -0.02 | 0.02 | -0.03 | -0.10 |
| Comm Services-Tilt | 0.87 | 0.81 | 0.85 | 0.87 | 0.86 | 0.82 | 0.76 | 0.60 | 0.93 | 0.77 | 0.63 | 0.29 | -0.09 | 0.20 | -0.13 |
| Consumer Stap-Tilt | 0.79 | 0.77 | 0.69 | 0.75 | 0.79 | 0.74 | 0.75 | 0.73 | 0.70 | 0.96 | 0.51 | 0.19 | 0.06 | 0.11 | -0.08 |
| Energy-Tilt | 0.71 | 0.62 | 0.65 | 0.67 | 0.69 | 0.74 | 0.56 | 0.44 | 0.63 | 0.57 | 0.95 | 0.32 | -0.01 | 0.22 | -0.06 |
| BP-Tilt | 0.45 | 0.25 | 0.28 | 0.44 | 0.41 | 0.43 | 0.22 | 0.03 | 0.35 | 0.23 | 0.40 | 0.82 | 0.33 | 0.38 | -0.22 |
| DY-Tilt | -0.02 | -0.10 | -0.25 | -0.13 | -0.06 | -0.04 | -0.09 | -0.02 | -0.14 | 0.03 | -0.07 | 0.37 | 0.83 | 0.22 | -0.12 |
| Size-Tilt | 0.25 | 0.17 | 0.14 | 0.27 | 0.17 | 0.24 | 0.17 | 0.02 | 0.21 | 0.14 | 0.19 | 0.29 | 0.21 | 0.82 | -0.16 |
| Ret_3M-Tilt | -0.04 | -0.02 | 0.01 | -0.02 | 0.00 | -0.02 | -0.01 | 0.01 | -0.05 | -0.03 | 0.05 | -0.19 | -0.13 | -0.11 | 0.64 |

Notes: This matrix shows the correlations between the 15 factor-mimicking portfolio returns and the 15 cross-sectional factor returns.

tilts exceed 93% correlations, and fundamental tilts achieve at least 82% (except Ret_3M at 64%), far above their non-target correlations.

Thus, MTE, as a mimicking principle, is highly effective: when a factor is not directly investible, an investor can simply express a view, and the MTE-based FM technique automatically generates a portfolio with dominant exposure to that factor, maximising the chance of capturing its performance.

Observation 4.1 (Factor-Mimicking (FM) Efficiency) *The MTE-based FM portfolio achieves sharp, pure analytical exposure and high out-of-sample correlation with its target factor.*

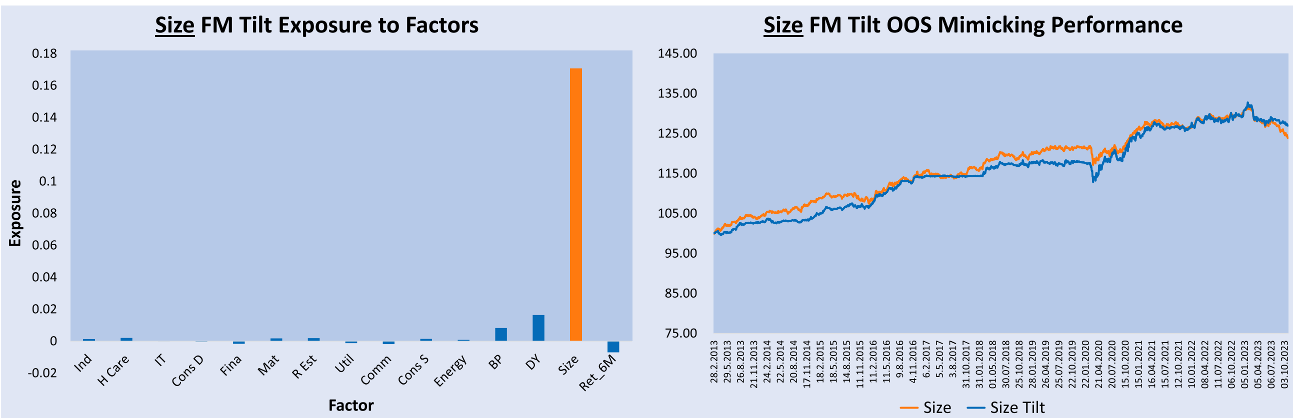


Figure 4. Size FM tilt exposures and OOS mimicking performance.

REMARK This confirms the FM technique effectively mimics factor performance, supporting Hypothesis 4.1 and validating Proposition 4.1 (Principle 1: MTE-Guided Mimicking) of Paper (I), providing a convenient and reliable way to gain maximum performance exposure to a target factor through view expression.

## 5. Testing MTE via SSM: idiosyncrasy of portable alpha

This section validates the MTE principle (Proposition 4.1, Paper I) by assessing how effectively MTE-based security-specific view portfolios (*SSM tilts*) mimic their target stocks while remaining market-insulated, using analytical exposure and OOS correlation.

### 5.1. Setup

Tests are conducted on two types of views: single-stock and stock-basket. The procedure is:

(1) Express a bullish *single-stock view* on a stock (e.g. 3M) at both the stock and stock-specific levels.†
(2) Using the SSM technique $(\mathbf{I} - \mathbf{BM}_{\mathrm{FM}})^{\mathrm{T}}$ and the October 2023 risk model, generate a portfolio on 01 Nov 2023. Assess the SSM tilt's weight and *return sensitivity* biases towards the stock, its common-specific risk split, and compare with the stock-level tilt to evaluate hedging of common risk without compromising target exposure.
(3) From 01 Mar 2013, apply the stock-specific view monthly using the preceding month's risk model to generate portfolios and compute OOS daily performance through to 02 Oct 2023 (128 months).
(4) Examine OOS correlations of the stock-specific tilt with the stock-level tilt and the market.
(5) Repeat for the *stock-basket view*.

Here, portfolio *return sensitivity* to stocks is defined as the composite impact of 1-standard deviation positive stock return shock on the subject portfolio's return, accounting for correlations between all stocks.

Since SSM is tested as a stand-alone technique here, the results provide general insights into the MTE-based hedging principle beyond the ABL model.

### 5.2. Results and analysis

**5.2.1. Single stock view.** Figure 5 displays the *stock-level* view tilt (upper panel) and its return sensitivity (lower panel). Since the view target is MMM, the tilt simply overweights MMM, reflecting natural investor behaviour. However, the lower panel shows that the tilt's return sensitivity is not solely to MMM due to stock correlations. Positive sensitivity to other stocks means a negative shock to any stock can adversely affect the tilt's performance.

† Tests on various stocks and baskets show SSM behaviour and MTE effectiveness are consistent across samples.

Table 7. The stock-specific view tilt minimises common risks.

| | Stock View Tilt | | Stock-Specific View Tilt | |
|---|---|---|---|---|
| | Variance | % | Variance | % |
| Common | 0.0008559 | 28.7% | 0.0000007 | 0.4% |
| Specific | 0.0021215 | 71.3% | 0.0001627 | 99.6% |
| Total | 0.0029774 | 100.0% | 0.0001634 | 100.0% |

Notes: Based on stock MMM, the breakdown of common vs specific variance in the stock-level view portfolio is compared to that of the stock specific-level view portfolio. Stock specific-level view portfolio has significantly lower common risk.

Table 8. The stock-specific view significantly reduces OOS correlation with the market.

| MMM Stock | OOS Correlation |
|---|---|
| Stock View Tilt vs the Market | 0.67 |
| Stock-Specific View Tilt vs the Market | − 0.11 |
| Stock-Specific View Tilt vs Stock View Tilt | 0.58 |

Notes: Based on stock MMM, this table shows that a stock-specific level view portfolio significantly reduces out-of-sample (OOS) correlation with the market compared to the same view expressed at the stock level.

Figure 6 presents the *stock-specific-level* view tilt (upper panel) and its return sensitivities (lower panel). While primarily focussed on MMM, the tilt also allocates long/short positions across the broader stock universe, which, in line with Hypothesis 4.3 (Paper I), serve to hedge MMM's common risk. The lower panel confirms that the SSM tilt is positively sensitive to MMM while *perfectly shielded* from other stocks, achieving 100% pure sensitivity and precise capture of the target stock's idiosyncratic performance without imposing efficiency-compromising constraints.

Table 7 compares the two portfolios' common versus specific variance. The stock-level tilt has 28.7% of variance from common risk, whereas the stock-specific tilt has only 0.4%, confirming the effectiveness of MTE-based common risk hedging (Hypothesis 4.3, Paper I).

Table 8 shows that the MMM stock view tilt has a 0.67 correlation with the market. While the MMM-specific tilt reduces market correlation to −0.11, it retains 0.58 correlation with MMM, demonstrating SSM's effectiveness in capturing stock-specific performance with minimal common risk.

**5.2.2. Basket view.** As with single-stock views, investors can express a view on an *investment theme* (a stock basket, Table 9) at either the basket or basket-specific level, each carrying different risk implications.

Figure 7 gives the *basket-level* view tilt (upper panel) and its return sensitivity (lower panel). While the tilt equally allocates to basket constituents, mirroring natural investor behaviour, stock correlations introduce noisy sensitivities, making the portfolio as responsive to random stocks as to the target basket and limiting capture of the theme's idiosyncratic performance.

Figure 8 displays the *basket-specific-level* view tilt (upper panel) and its return sensitivity (lower panel). The tilt focuses

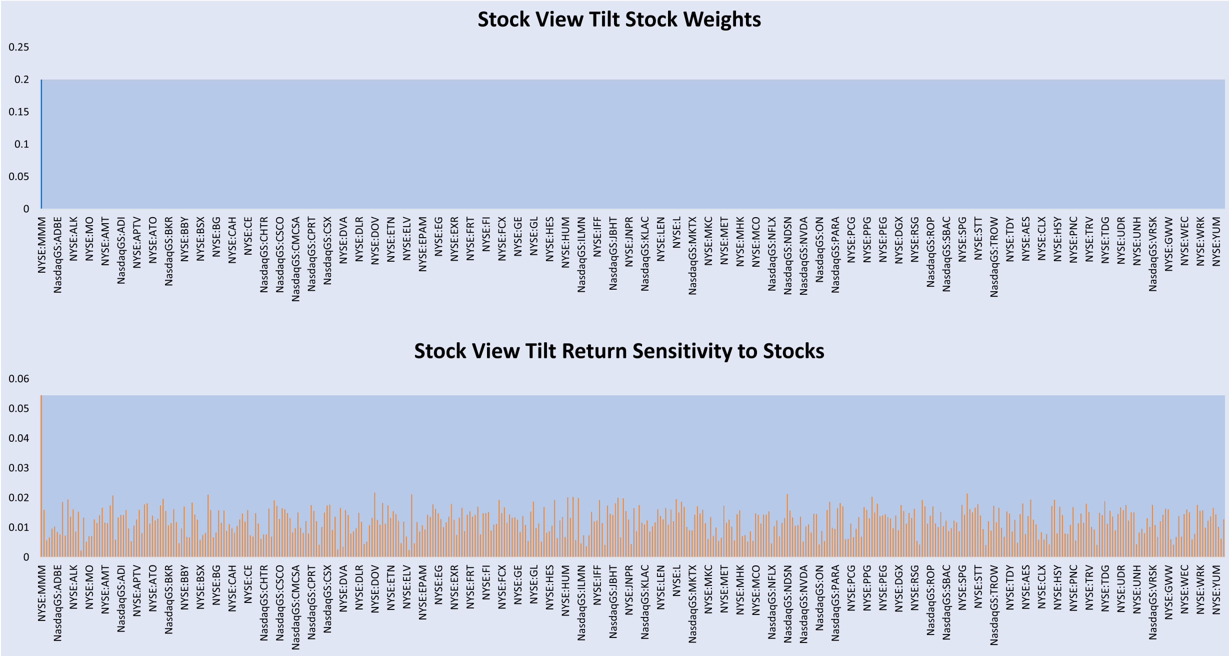


Figure 5. The stock-level view tilt and its sensitivity.
Based on stock MMM, the upper panel shows the stock-level view *portfolio weights*. The lower panel shows its composite *return sensitivity* to 1-standard deviation stock return shocks. In addition to the desired sensitivity to MMM, the undesirable noisy sensitivities mean any other stock movements can also impact the performance of our view portfolio.


Figure 6. The stock specific-level view tilt and its sensitivity.
Based on stock MMM, the upper panel shows the stock-level view *portfolio weights*. The lower panel shows its composite *return sensitivity* to 1-standard deviation stock return shocks. In addition to the desired sensitivity to MMM, the undesirable noisy sensitivities mean any other stock movements can also impact the performance of our view portfolio.

on basket constituents while SSM introduces long/short positions as a common risk hedge. Yet the sensitivity profile exhibits neat positive exposure only to the target basket, *perfectly insulated* from other stocks, enabling 100% pure exposure and precise capture of the theme's idiosyncratic performance without imposing efficiency-undermining optimiser constraints.

Table 10 presents that 77.2% of the basket-level tilt's variance comes from common risk, versus only 0.8% for the basket-specific tilt, confirming the effectiveness of MTE-based common risk hedging (Hypothesis 4.3, Paper I).

Table 11 illustrates the basket view tilt has a 0.86 correlation with the market. While the basket-specific tilt reduces market correlation to 0.21, it retains 0.62 correlation with

Table 9. The sample stock basket.

| Ticker | AAPL | AMD | AMZN | GOOGL | META | MSFT | NFLX | NVDA | QCOM | VZ | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Weight | 10% | 10% | 10% | 10% | 10% | 10% | 10% | 10% | 10% | 10% | 100% |

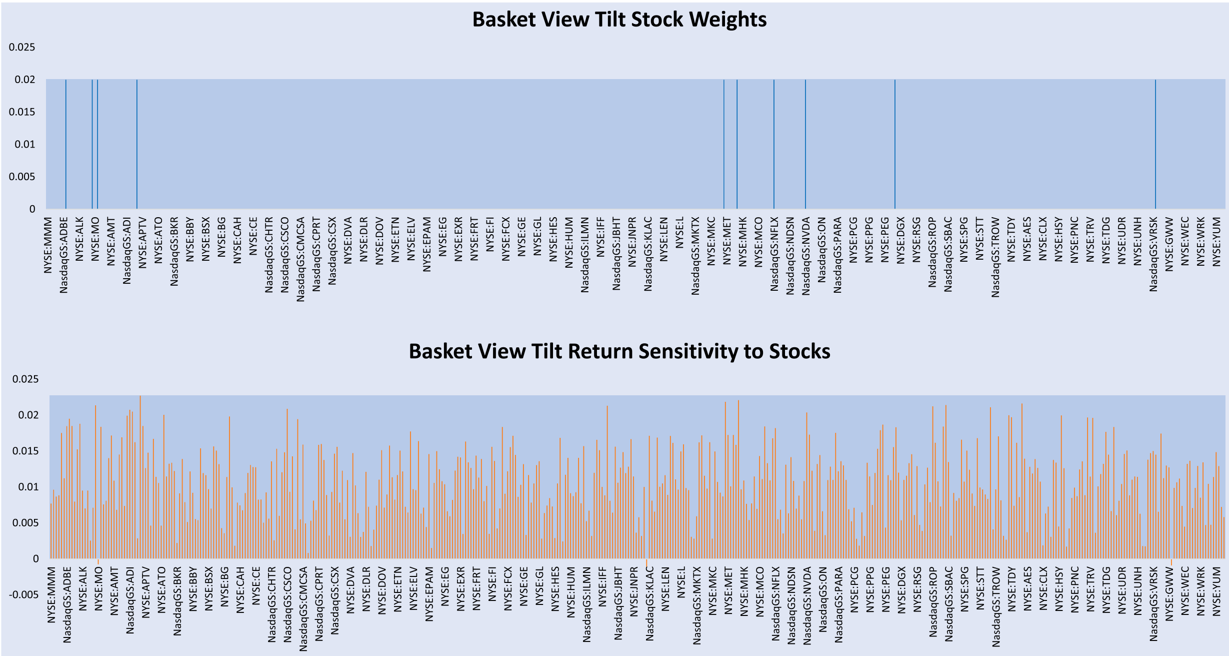


Figure 7. The basket-level view tilt and its sensitivity.
Based on the sample basket, the upper panel shows the basket-level view *portfolio weights*. The lower panel shows its composite *return sensitivity* to 1-standard deviation stock return shocks. In addition to the desired sensitivities to the basket, the undesirable noisy sensitivities mean any other stock movements can also impact the performance of our view portfolio.

Table 10. The basket-specific view tilt minimises common risks.

| | Basket View Tilt | | Basket-Specific View Tilt | |
|---|---|---|---|---|
| | Variance | % | Variance | % |
| Common | 0.0009376 | 77.2% | 0.0000005 | 1.8% |
| Specific | 0.0002774 | 22.8% | 0.0000255 | 98.2% |
| Total | 0.0012150 | 100.0% | 0.0000260 | 100.0% |

Notes: Based on the sample basket, the breakdown of common vs specific variance in the basket-level view portfolio is compared to that of the basket specific-level view portfolio. Basket specific-level view portfolio has significantly lower common risk.

Table 11. The basket-specific view significantly reduces OOS correlation with the market.

| Basket | OOS Correlation |
|---|---|
| Basket View Tilt vs the Market | 0.86 |
| Basket-Specific View Tilt vs the Market | 0.21 |
| Basket-Specific View Tilt vs Basket View Tilt | 0.62 |

Notes: Based on the sample basket, this table shows that a basket-specific level view portfolio significantly reduces out-of-sample (OOS) correlation with the market compared to the same view expressed at the basket level.

the target basket, demonstrating SSM's effectiveness in capturing basket-specific performance with minimal common risk.

Thus, MTE, as a hedging principle, is highly effective: to gain an idiosyncratic exposure, an investor can simply express a view on the security-specific return, and the MTE-based hedging technique automatically generates a portable alpha with dominant exposure to the target while hedging common factor risks, thereby maximising the chance of capturing its idiosyncratic performance.

Observation 5.1 (Security-Specific Mimicking (SSM) Efficiency) *The MTE-based security-specific view tilt (portable alpha) achieves sharp, pure analytical exposure and high out-of-sample correlation with its target security or basket, while minimising common risk.*

Remark This confirms that SSM effectively mimics security-specific performance with built-in common-risk hedging, supporting Hypotheses 4.2 and 4.3 and validating Proposition 4.1 of Paper I. It provides a convenient, reliable method to construct portable alpha with pure idiosyncratic exposure via view expression.

## 6. Implications

Validating SAR and MTE as guiding principles simplifies and clarifies *active* portfolio selection, unifies theory and practice, and addresses the seven portfolio optimisation challenges identified by Salo *et al.* (2024, pp. 13–14):

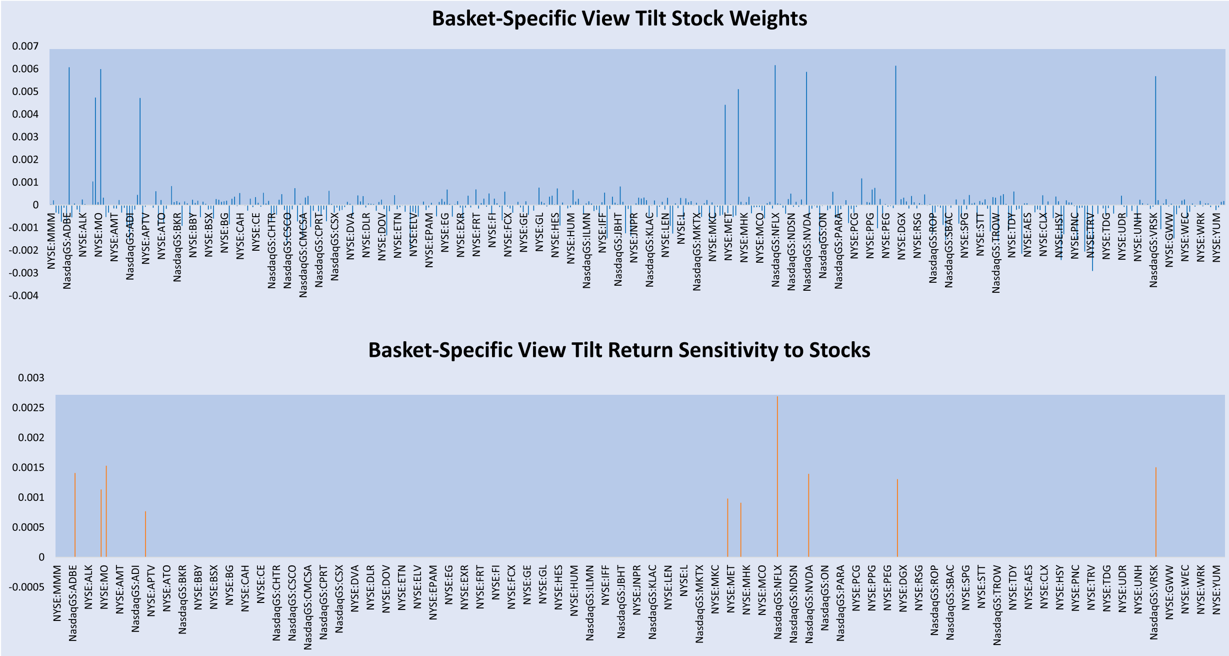


Figure 8. The basket specific-level view tilt and its sensitivity.
Based on the sample basket, the upper panel shows the basket specific-level view *portfolio weights*. The lower panel shows its composite *return sensitivity* to 1-standard deviation stock return shocks. Aside from the desired sensitivities to the basket, the view portfolio becomes perfectly insulated from the influences of all other stocks in the universe.

**Broader Span of Criteria and Uses**. SAR and MTE internally blend unlimited objectives through simple and intuitive view expressions, removing the need for complex, rigid objective settings that can quickly become intractable with multiple objectives.

**Non-Numerical Data Sources**. SAR incorporates qualitative, ambiguous, and subjective information as effectively as quantitative data, leveraging ambiguity to enhance performance (Proposition 4.2, Paper I; Subsection 3.2.2; Observations 3.1, 3.2).

**Subjective Perceptions of Uncertainties**. Subjective treatments of the covariance matrix, including view uncertainty and correlation adjustments, is supported through portfolio massage and can enhance performance beyond MPT (Observation 3.2).

**Optimising Portfolio Decision Processes**. MPT's passive nature dilutes conviction, whereas SAR and MTE optimise decision processes by aligning scientific portfolio selection with investor cognition, enabling *intentional* exposures (Observations 3.1, 4.1, 5.1; Proposition 4.2, Paper I).

**Robustness**. Unlike conservative robust optimisation, portfolio massage leverages uncertainties to proactively reconcile *tactical* views with *strategic* goals, providing robust error management that can outperform traditional methods (Observation 3.2; Proposition 4.2, Paper I).

**Dynamic Monitoring, Discovery and Adapting**. Natural investor views are accurately mapped to portfolios (Theorem 4.1, Paper I), with analytics revealing and correcting unintended exposures through portfolio massage, without any loss of efficiency.

**Enhancing Human Interactions**. The framework enables decision-makers to exchange and negotiate stakeholder views in meetings by dynamically blending them into a portfolio with real-time analytics-based feedback on-screen, ensuring transparent and efficient decision-making.

## 7. Conclusion

This Paper (II), the second of a two-part series, empirically validates the *Subjective Allocation Rule* (SAR) and the *Minimum Tracking Error* (MTE) as the two efficiency principles underlying *active* portfolio-selection (PS) theory and practie. **Skill-controlled view simulation** (SVS) tests confirm that SAR generates alpha (outperformance) proportional to investors' forecasting *skill*, while **comparative subjectivity tests** (CST) show that investors' *subjectivity* can enhance – rather than hinder – performance, outperforming traditional mean-variance (MV) portfolios. Factor- and security-specific mimicking tests further demonstrate that MTE enables investors to harness a broad range of information sources effectively. Together, SAR and MTE form a cognitively aligned *Bayesian Allocation Framework* that translates natural and diverse investor beliefs into the purest active portfolios, most precisely reflecting their genuine intent.

Practically, this framework enables an ideal division of labour: humans drive the *art* of view formulation (intent discovery), while the framework ensures the *sciencific efficiency* of portfolio implementation and feedback. Investors of any sophistication can express their directional or quantitative beliefs on familiar subjects of their choice, whether directly tradable or not, and let the framework translate them into portfolios with analytics-based feedback. *Analytics-guided portfolio massage* then leads them through the discovery of their true intent – where rationale is clarified, ambiguities resolved, and the portfolio naturally selected.

This paper makes four key contributions: (1) *Empirical Validation*: Systematically validates SAR and MTE as efficient, alpha-generating, and cognitively friendly principles for active portfolio selection. (2) *Methodologies*: Introduces SVS to isolate PS models from alpha models, avoiding *joint-test* biases; and a novel CST framework to assess the impact of subjectivity on performance. (3) *Blending Information*: Demonstrates seamless integration of qualitative and subjective inputs into portfolio optimisation. (4) *Performance Drivers*: Shows that appropriate investor subjectivity can complement forecasting skill to enhance portfolio outcomes.

Looking ahead, Papers (I) and (II) collectively lay the foundation for broader methodological and theoretical synthesis. First, empirical and theoretical work can explore the conditions under which subjective interventions become essential for mitigating estimation errors and generating performance breakthroughs (see e.g. Cheung 2024b). Second, Cheung (2026b) tests the remaining part of Lemma 4.2 of Paper (I) – namely, that the *IR scales with betting size* – pointing towards a new Fundamental Law of Active Management that endogenises portfolio *concentration*, as exemplified by investors such as Warren Buffett and Bill Ackman, in contrast to the *breadth* of Grinold and Kahn (1999). Third, to achieve full theoretical closure, Cheung (2026a) develops a meta-theoretical approach to validating the Bayesian Allocation Framework as a Unified Portfolio Theory, capable of coherently explaining, integrating, and guiding with parsimony a broad range of heterogeneous real-world practices, including diverse heuristics, folk wisdom, and more. Together, these efforts aim to further reconcile PS theory and practice, offering both academics and practitioners general yet simplified guidance to navigate portfolio decisions in a complex and uncertain world.

Beyond these foundational advancements, the robustness of SAR and MTE can be further stress-tested across diverse asset classes, market regimes, and alternative heuristics. Rather than increasing model complexity, this parsimonious framework may provide a path to reconcile non-normality, psychological biases, and machine learning-generated signals within a unified structure. On the practical front, the transition from theory to toolset – via AI-assisted decision-support and interactive 'portfolio massage' – seeks to operationalise these principles, demonstrating that a scientific portfolio theory can be as agile in practice as it is rigorous in its foundations.

Overall, the Bayesian Allocation Framework, formulated in Paper (I) and validated here, reframes active portfolio selection as a form of *view-driven, feedback-responsive* goal-discovery rationality, moving beyond purely prescriptive, normative approaches towards a paradigm that captures the dynamic interplay between human cognition, information heterogeneity, and market complexity. By doing so, it lays a robust foundation for further innovation in both the theory and practice of active portfolio management.

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## Author contributions

CRediT: **Wing Cheung:** Writing – original draft, Writing – review & editing

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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## Appendix. more backtesting results

### *A.1. The factor mimicking (FM) technique backtesting*

Figures A1–A13 visualise the additional FM technique testing results against the other 13 risk model factors.

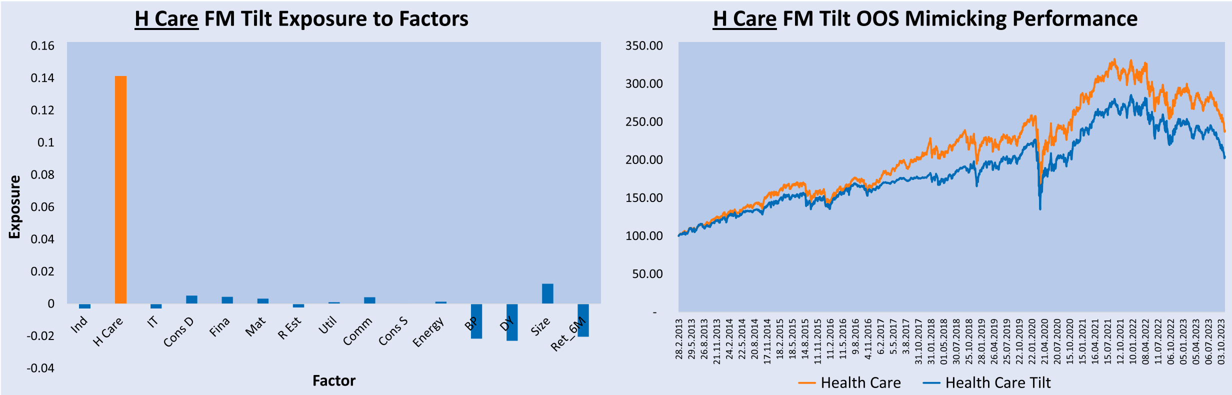


Figure A1. Health Care FM Tilt exposures and OOS mimicking performance.

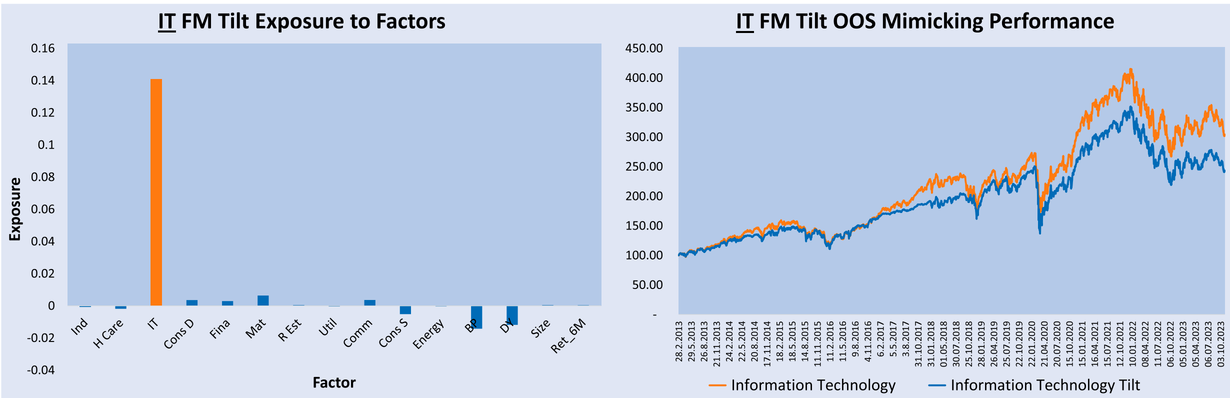


Figure A2. Information Technology FM portfolio exposures and OOS performance.

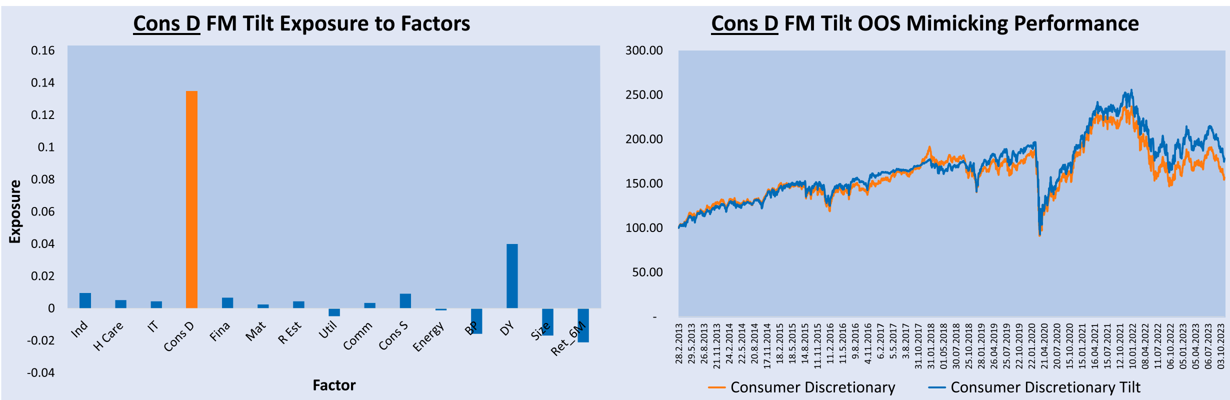


Figure A3. Consumer Discretionary FM portfolio exposures and OOS performance.

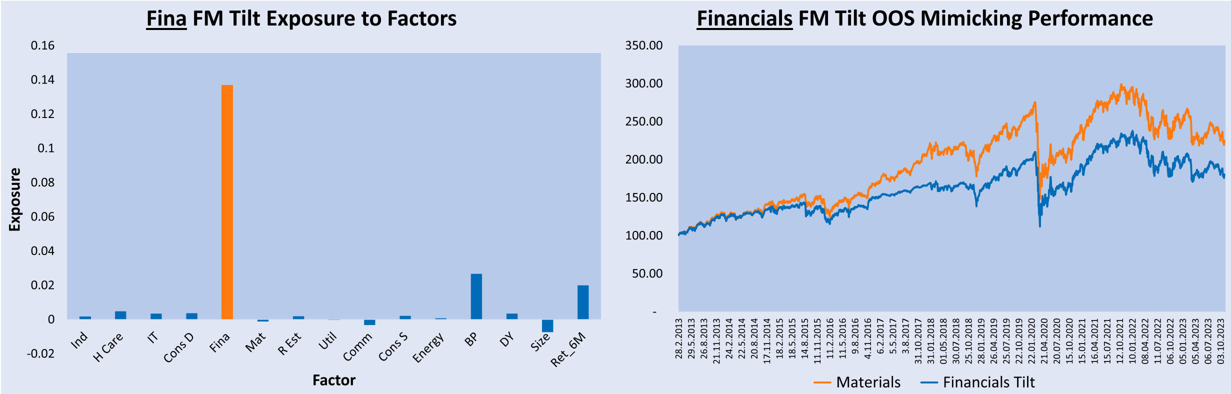


Figure A4. Financials FM Tilt exposures and OOS mimicking performance.

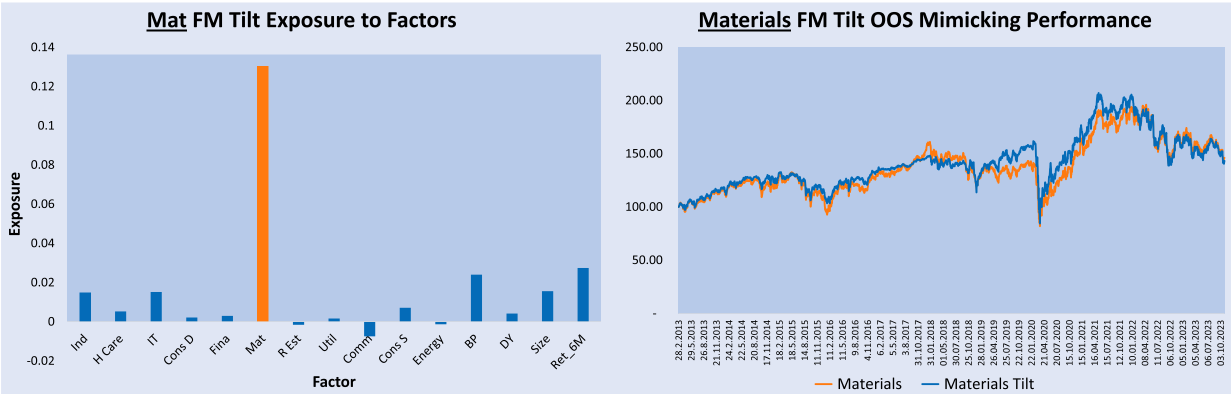


Figure A5. Materials FM Tilt exposures and OOS mimicking performance.

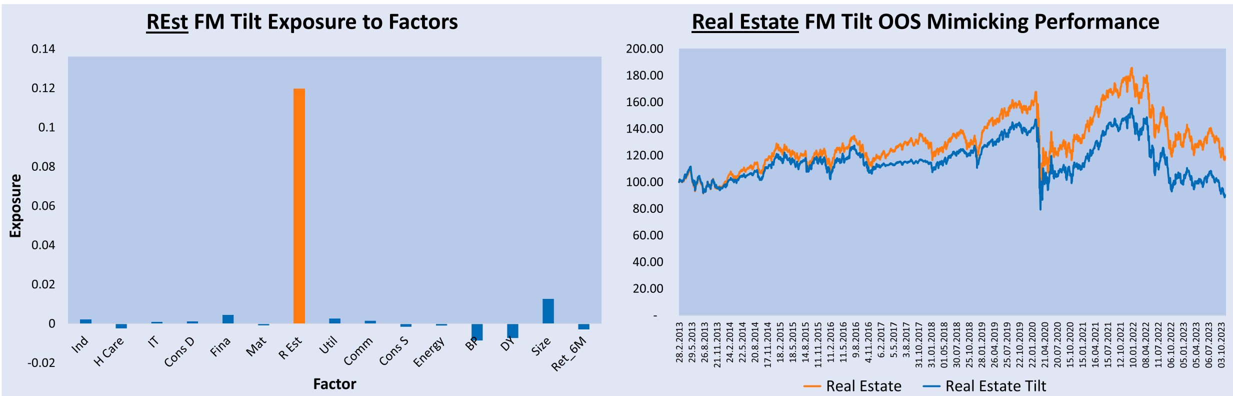


Figure A6. Real Estate FM Tilt exposures and OOS mimicking performance.

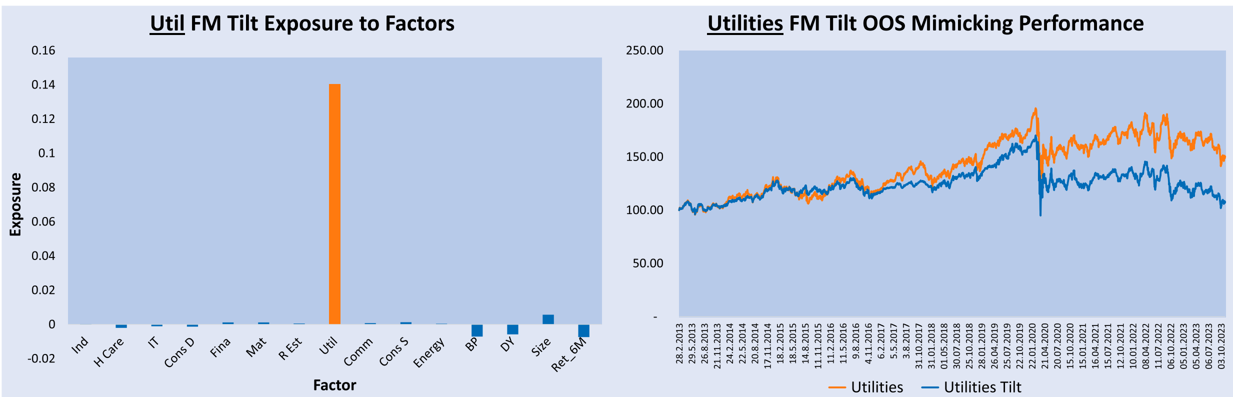


Figure A7. Utilities FM Tilt exposures and OOS mimicking performance.

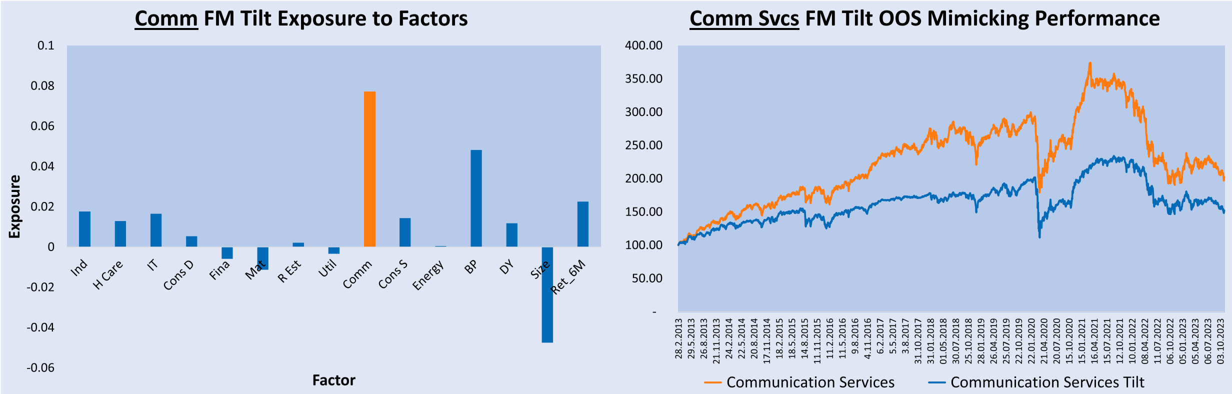


Figure A8. Communication Services FM portfolio exposures and OOS performance.

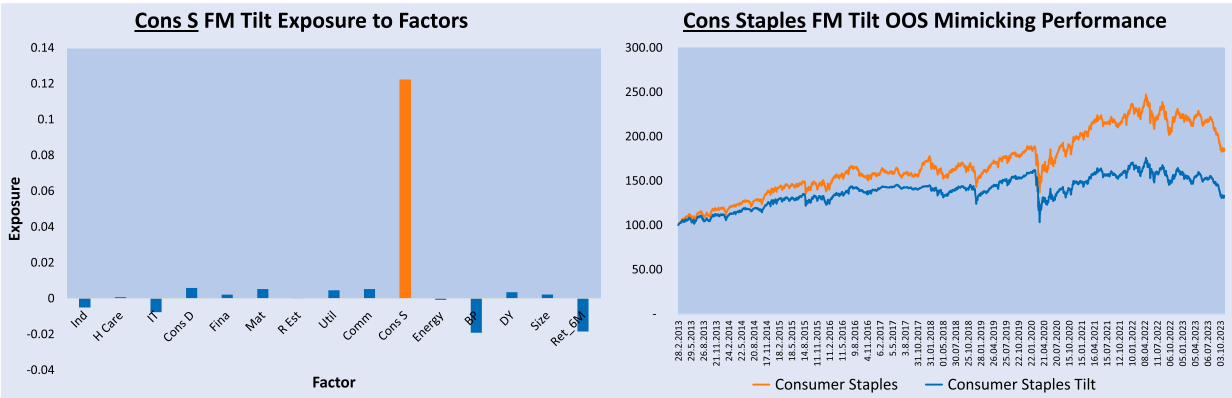


Figure A9. Consumer Staples FM Tilt exposures and OOS mimicking performance.

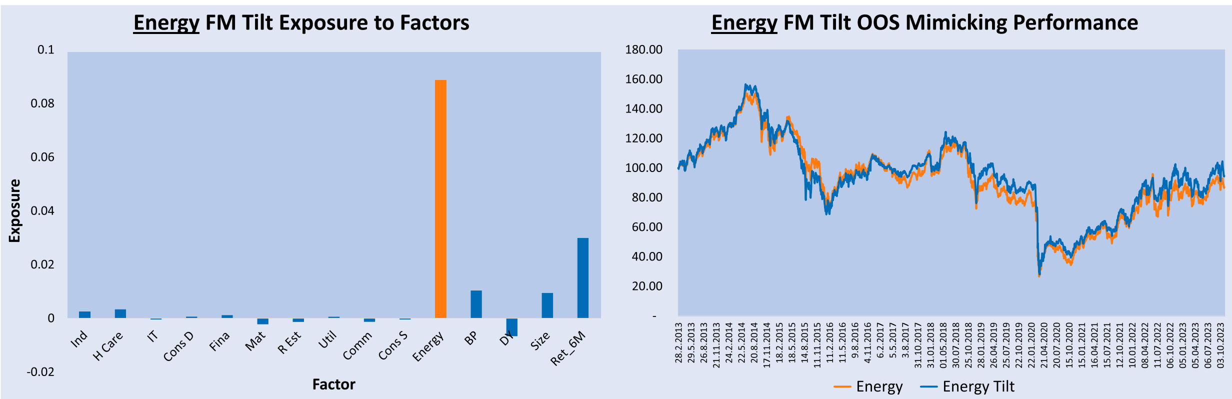


Figure A10. Energy FM Tilt exposures and OOS mimicking performance.

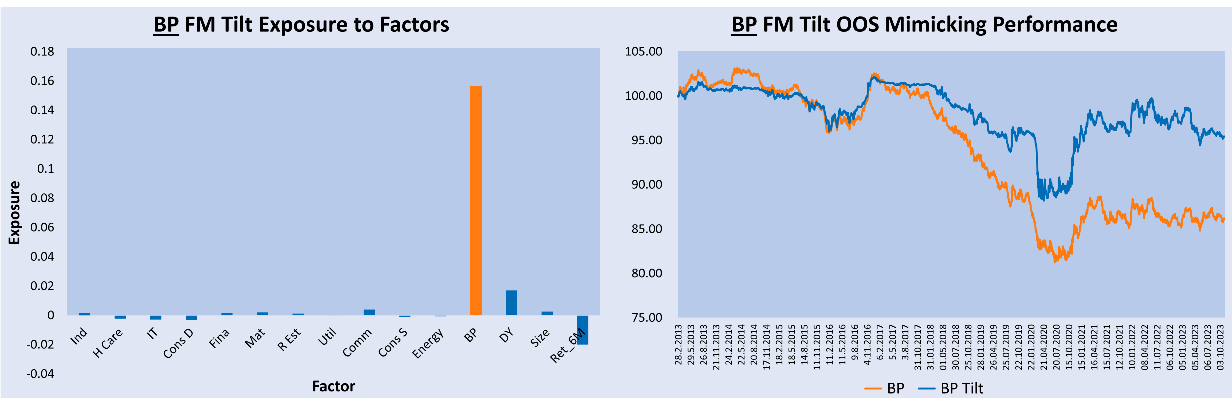


Figure A11. BP FM Tilt exposures and OOS mimicking performance.

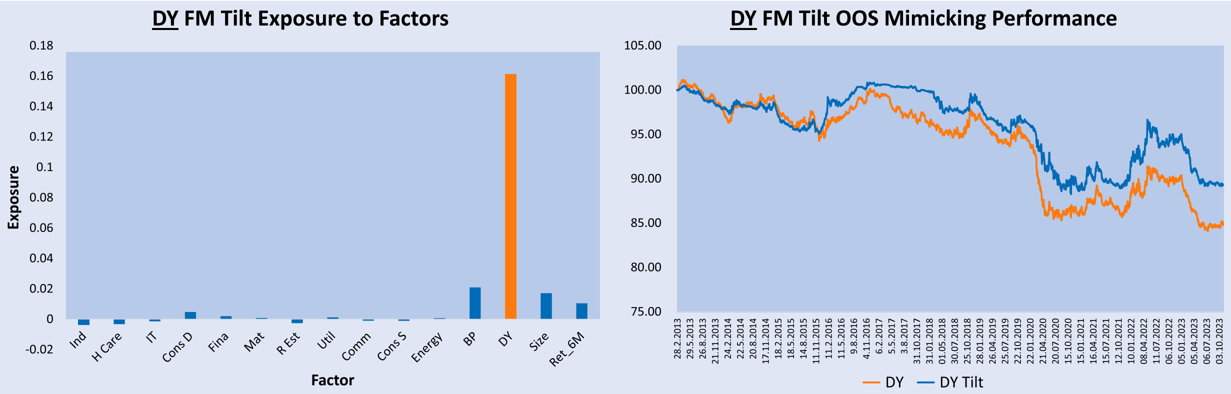


Figure A12. DY FM Tilt exposures and OOS mimicking performance.

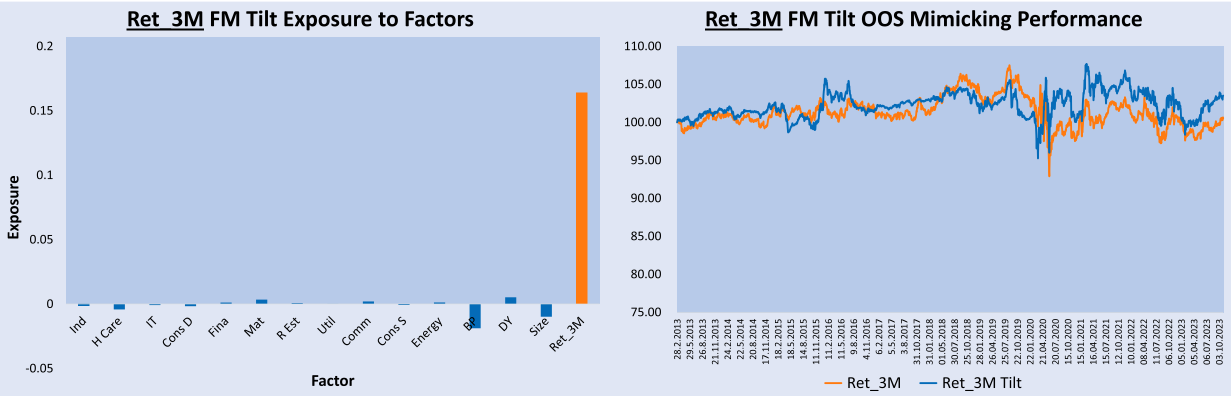


Figure A13. Ret_3M FM Tilt exposures and OOS mimicking performance.